Darker Blue: How Small Donors Drive Congressional Polarization

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Abstract

Over the previous two decades, campaign contributions from small donors to congressional elections have surged, and new technologies have transformed the process of political fundraising. How does money from small donors change how legislators take positions and behave in Congress? ActBlue is a digital platform that has lowered the non-monetary costs of individual contributions and has processed billions of dollars in donations for Democratic candidates and causes. Using a difference-in-differences design based on staggered ActBlue adoption by House candidates, I find that when candidates join ActBlue, they raise more money from individual donors. This is predominantly driven by growth in small-donor contributions. The source of contributions to these candidates shifts to the ideological left. Those elected to Congress tend to behave more liberally across multiple metrics, relative to their own previous behavior. These findings suggest that pursuit of small and individual donors contributes to congressional polarization.

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1 Introduction

How do campaign contributions shape political outcomes and contribute to political polarization? The cost of Congressional elections has surged in past years, and fundraising is a critical component of running for office. Researchers, policymakers, and individuals often turn to money in politics to explain polarization and policy outcomes in the United States. Concerns about responsiveness to donors have led researchers to study contributions from access-seeking groups and political insiders (Barber, 2016; Bonica, 2020; Desmarais et al., 2015; Fouirnaies & Fowler, 2022; Fouirnaies & Hall, 2018; Fowler et al., 2020; Grier et al., 2023; Powell & Grimmer, 2016), but these types of donors are only a portion of candidates' donor pools. As contributions from smaller donors have risen in recent years, it is increasingly important to understand whether legislators are responsive to small donors. The correlation between small donor contributions and legislator extremity is well-established: legislators who raise more from small donors tend to be more ideologically extreme (Albert & La Raja, 2020; Culberson et al., 2019; La Raja & Schaffner, 2015). Do donors drive ideological extremity? The prevailing mechanism within the literature is that individual and small donors, who themselves are more extreme than the electorate, prefer supporting non-moderate candidates (Barber et al., 2017; Keena and Knight-Finley, 2019; Meisels et al., 2022, but see Meisels, 2022). Extreme candidates, aided by these donors, then enter office (Barber, 2016; Kujala, 2020). Because the supply of political donations is tied to candidate extremity, and because there is little political appetite to constrain the participation of small donors (unlike the case of corporations and PACs), it is difficult to establish whether money from small donors directly pushes legislators toward ideological and partisan extremes.

In this paper, I find that when Democratic candidates receive a larger share of contributions from smaller donors, their donor pools become more liberal, and their legislative behavior shifts to the left. This paper is the first to show that money from small donors causes legislators to become more polarized, and provides some of the most direct evidence on the impact of money on legislative behavior. I use ActBlue adoption among Democratic candidates running in federal House general elections as a shock in order to examine how small donor involvement impacts overall donor pool ideology and legislative behavior. ActBlue is the preeminent website to process payments for Democratic candidates and progressive causes. It combines tools for effective fundraising, regulatory compliance, and donor data management for candidates, as well as lowers the non-monetary costs of making a political contribution for donors. ActBlue adoption provides an ideal case for studying the effects of small donors on legislative behavior, as the platform encourages contributions from small donors and has been widely embraced by Democrats in Congress in recent years.

I introduce a new dataset on ActBlue adoption among general election House candidates within the Democratic party. I combine this dataset with existing data on candidate and district attributes and fundraising metrics. Using a two-way fixed effects (legislator and time) difference-in-differences strategy based on staggered ActBlue adoption among House candidates, I find that when candidates join ActBlue, contributions from non-large individual donors (those giving under $1000 per candidate) substantially increase, primarily driven by unitemized donors, who give less than $200 to candidates. Smaller donors become a larger share of candidates' fundraising portfolios. When candidates start fundraising with ActBlue, the ideological balance of their donors also shifts to the left. Once in office, legislators who raise money on ActBlue position themselves and vote more liberally on key issues, evidenced by leftward shifts in their interest group-based ideology scores. Ideology scores based on cumulative voting records move to the left on average, as candidates on ActBlue tend to vote more with the majority of Democratic legislators and more against the majority of Republican legislators.

This article proceeds as follows. First, I describe ActBlue's role in the fundraising process and discuss how money from small donors brought about by ActBlue may spark donor and congressional polarization. The data section describes the data collection process, empirical methods, ActBlue adoption, and possible issues for causal inference. In the results section, I investigate candidates' financial returns to ActBlue adoption, by donor type. I show that candidates see the largest increases in contributions from small donors, and that smaller donors become a larger component in candidates' fundraising portfolios. I then examine the interaction between ActBlue adoption and the ideological bend of congressional donors. Finally, the paper investigates whether congressional behavior moves leftward when House members use ActBlue. I briefly consider whether WinRed, the Republican analog to ActBlue, brings about similar changes. The concluding section contextualizes these results against the broader backdrop of trends in fundraising, donor ideology, and congressional behavior.

1.1 ActBlue technology enables candidates to effectively reach smaller individual donors

ActBlue, founded in 2004, is an online platform that facilitates contributions to Democratic and liberal candidates and causes. Most Democratic congressional candidates from the early 2000s raised money on the internet through firms like PayPal, Google, or NGP, and throughout the late 2000s to mid-2010s, slowly joined the ActBlue platform.[1] ActBlue altered how individuals make campaign finance transactions by making it easier for prospective donors to give and re-give. The platform increased the types of payments accepted and enabled individuals to schedule recurring contributions. Creating an ActBlue Express account enabled prospective donors to give in a single click, bypassing the need to input payment or employment information. The group has also invested in maintaining and testing a modern aesthetic and simple, mobile-compatible user experiences, to "make [the contributing process] as frictionless as possible," according to ActBlue's executive director (Reilly, 2019).

In addition to making it easier to donate once a prospective donor lands on the ActBlue site, ActBlue enables campaigns to create more opportunities for donors to land on a

[1]Observation based on original data collection

candidate's fundraising portal. In both email and advertisement contexts, campaigns can use ActBlue tools to A/B test which messages effectively convert prospective donors into donors. The organization conducts experiments and reports its findings to campaigns, enabling campaigns to skip over ineffective solicitation strategies (Willis, 2014). Furthermore, ActBlue provides tools for campaigns to identify and keep tabs on potential donors.

These innovations center around converting prospective donors into donors and current donors into recurring donors. Since the largest donors are bound by FEC contribution limits, there is little ActBlue can do to aid campaigns in squeezing more money out of larger donors. Since larger donors are more likely to be donating for access-seeking reasons, the impersonal nature of making an online contribution is additionally unappealing, relative to attending an in-person fundraising even with the opportunity to network with the candidate.

Finally, ActBlue also expands the universe of candidates who can use sophisticated fundraising tools. ActBlue provides its services to campaigns by taking a fixed percentage of contributions, whereas access to similarly advanced tools traditionally required contract-based payment. For campaigns without substantial cash reserves, entering into a contract with upfront costs is often not financially feasible. Nor can lower-profile campaigns employ or regularly consult with marketing professionals or data analysts. James Cargas, a candidate from Texas, explained[2] that if not for ActBlue, campaigns without war chests would likely have turned to other solutions without upfront costs, like PayPal or credit card processing through a local bank. Without ActBlue, candidates would still be able to fundraise online, but the Democratic online fundraising landscape would have remained decentralized for a longer period, with fewer innovative tools available to all candidates, especially those unable to raise substantial sums of early money. Taken together, ActBlue enables more campaigns to identify and engage with more prospective donors, more frequently. Notably, these innovations center around smaller individual donors.

[2] In conversation with the author

1.2 Smaller donors and political ideology

Candidates have strong incentives to adopt fundraising strategies that reliably attract donations. Over the past two decades, both the composition of the donor pool and the cost of running for office have changed dramatically. By 2020, the average Democratic congressional general election candidate raised over $4 million, nearly four times what candidates raised two decades earlier.[3] These rising costs can shape who chooses to run (Carnes, 2018) and affect electoral outcomes (Thomsen, 2023). As campaigns become more expensive, electoral competition degrades and incumbents tend to be more advantaged (Fouirnaies, 2021), though the marginal returns on spending diminish slower for challengers – if they can raise enough money (Bonneau & Cann, 2011). Given the centrality of fundraising to electoral viability, candidates have little reason to avoid small donor money when it becomes accessible.

What happens when candidates raise more from small donors? Prior work suggests that an increase in small-dollar fundraising may coincide with greater ideological extremity among both donors and candidates (La Raja, 2014; La Raja & Schaffner, 2015; Pildes, 2019). This expectation is grounded in two key findings. The first finding is on small donor preferences. Smaller donors are ideologically similar to larger individual donors (Albert & La Raja, 2020; Alvarez et al., 2020), who are more ideologically extreme than both access-seeking PACs (Barber, 2016) and voters (Bafumi & Herron, 2010; Broockman & Malhotra, 2020). Additionally, small donors do not shy away from ideologically extreme candidates (Albert & La Raja, 2020; Culberson et al., 2019; Meisels et al., 2022), and candidates with extreme reputations, like Marjorie Taylor-Greene and Alexandria Ocasio-Cortez, have disproportionately high small donor support. The second finding concerns responsiveness to donors. Access-seeking PACs appear to moderate legislative behavior (Barber, 2016) and legislators are responsive to both individual and group-based donors (Canes-Wrone & Miller, 2021; Grier et al., 2023; Kalla & Broockman, 2016; Kujala, 2020; Stratmann, 1995). With these results in mind, we should expect that an increase in money from smaller donors will

[3] https://www.opensecrets.org/elections-overview?

elicit a leftward ideological response from legislators.

Still, these patterns may not hold in the context of ActBlue. The platform may not substantially increase small-donor participation, or the small donors it attracts may differ from those traditionally associated with ideological pressure. For instance, high-profile but uncompetitive Democratic challengers have drawn millions from small donors, raising concerns about the efficiency of Democratic donor behavior (Hersh, 2020; Sokolove, 2022). Small donors brought in via new technologies may diverge from the ideologically intense archetype or may not push candidates in any ideological direction. Indeed, recent evidence finds ActBlue donors differ materially from larger individual donors (Bouton et al., 2022). Alternatively, any ideological pressure from small donors might be offset by moderating forces such as PACs, resulting in little net effect on candidate behavior. And causality may run in only one direction: ideologically extreme candidates may attract small donors, without altering their own positions in response. This paper analyzes small donor participation and the ideological composition of the donorate. A key contribution is the focus on within-legislator ideological change, which demonstrates that legislators respond to small donors, not just the reverse.

Even where patterns of polarization do emerge following ActBlue adoption, they may result from broader changes in donor composition rather than just increases in the quantity of smaller donors. Because internet campaigning enables donors to learn and give to candidates beyond their congressional district, it is plausible that results actually stem from out-of-state, nationalized donor pools. Others (Baker, 2016; Canes-Wrone & Miller, 2021) have shown that legislators who are more reliant on out-of-district donors are more likely to side with their party's national donors. I discuss out-of-state donors, serial donors, and new donors in Appendix Table A14, and find that ActBlue adoption does not increase participation among these other types of donors.

ActBlue is particularly adept at encouraging contributions from a certain type of donor

– typically low-dollar, high-frequency, ideologically motivated individuals – who happen to be most prevalent at the lower end of the contribution distribution. While "small donor" often refers to a technical definition of an unitemized donor, applicable to donors give less than $200 per election,[4] the mechanisms that I describe in this paper are not confined to this threshold. Rather, they reflect a behavioral pattern that is strongest among unitemized donors, but extends to larger individual contributors as well: I show that individual donors giving $200-$1000 exhibit weaker but directionally similar trends as unitemized donors.

2 Data and Methods

I used the Database on Ideology, Money in Politics, and Elections (DIME) from Bonica (2014) to obtain candidate congressional district, CF Scores, itemized contributors, and incumbency status. I used OpenSecrets datasets on the sum and share of contributions raised by candidates from donors unitemized donors.[5] Some general election candidates are missing from the OpenSecrets data. For these candidates, I manually recorded their receipt totals and their totals of unitemized contributions from the FEC website or verified that they did not raise enough funds to report data to the FEC. Because Senate elections are comparatively infrequent, I subset the data to only House elections. I used data from 2002-2016.[6] To ensure the focus on candidates who are actively fundraising, I restricted the sample to candidates raising over $20,000 in a given cycle.

I appended additional data on candidates and their congressional districts. I used interest group ratings from Vote Smart and roll call voting data or scalings from Clinton et al. (2004), Fowler (2024), and Lewis et al. (2023). Data on legislator and candidate ideology

[4]Under this definition, primary and general elections are two separate elections.

[5]In theory, this can lead to some strange labeling decisions. If a single donor gave $199 to 200 candidates, they are a "small" donor, and if a single donor gave four contributions of $50 to one candidate, they would be labeled as a non-small donor. However, the latter behavior is far more common than the former, and most donors give once.

[6]By 2018, ActBlue essentially gained market dominance, and few serious new candidates chose to use another platform.

are described in more detail alongside Tables 5-6. For regressions with controls, I used Daily Kos calculations on presidential Democratic vote share by congressional district to calculate district safety.

2.1 ActBlue adoption

Measuring ActBlue adoption is not necessarily straightforward. I consider a candidate to be an ActBlue adopter when they use ActBlue to process online contributions originating from their campaign website. To determine whether a candidate encouraged prospective donors into the ActBlue environment, I used the Internet Archive's Wayback Machine and the Library of Congress' archive of candidate websites. After obtaining a candidate's website URL for a given campaign year, I navigate to the website's contribution page. From there, it is usually clear what service the candidate uses to process payments.

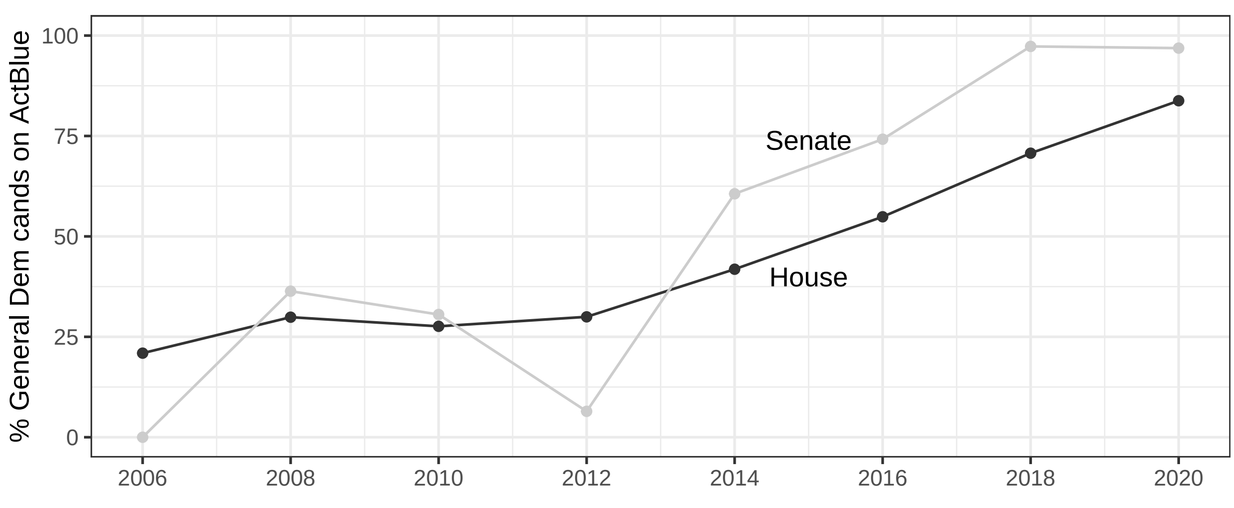
There is some uncertainty in this data-collection process, as some donation pages and some complete websites were never fully captured by the archives. Additionally, not all candidates had websites or solicited donations on their websites. Candidates in districts with little electoral competition were among the last to build websites, and long-term incumbents with robust offline contribution networks, such as Rep. Nancy Pelosi, were among the last to solicit contributions on their websites. I code these uncertain instances (about 8% of observations) as non-adopters.

The website-based strategy is not without drawbacks. Many candidates actively solicited contributions via ActBlue (through emails, texts, or ads) but directed website visitors to other payment processors. For example, Hillary Clinton's 2016 presidential campaign website brought donors to NGP's FastAction, while still hosting a landing page (and raising money) through ActBlue. Earlier in ActBlue's history, supporters even raised money for candidates without the candidate's knowledge of the platform. I code candidates as adopters only when their campaign websites redirect visitors to ActBlue, ensuring that adoption reflects an active campaign decision. This approach avoids false positives that would arise from relying solely on FEC data, where candidates might passively receive ActBlue funds. However, it also means that some known users (including some raising over $100k via the platform) are coded as non-adopters if their websites did not redirect to the platform, potentially attenuating treatment effects in models with candidate-specific trends and leads, which I discuss in following sections.

2.1.1 When did candidates join ActBlue?

The vast majority of Democratic general election candidates now use ActBlue as their primary online contributions processor. However, in ActBlue's early days, candidates wanting to fundraise online had many competitors to choose from. Unlike its Republican counterpart, WinRed, where Republicans quickly coalesced on a single platform endorsed by party leadership (S.-Y. S. Kim & Li, 2025), Figure 1 shows that ActBlue convergence was took years.

Figure 1 – ActBlue adoption among general election Democratic candidates over time, by chamber



What factors do campaigns cite for choosing ActBlue? Sam Spencer,[7] who has worked

[7] in conversation with the author

in key finance roles in Democratic and left-leaning campaigns across multiple levels of government, characterizes the movement towards ActBlue as multi-faceted, with inertia and user-friendliness for both campaign workers and donors serving as major reasons that campaigns used the platform. When running for House in Texas, James Cargas[8] decided to use ActBlue because he had previous experience with the platform as a treasurer for state PACs, and that the platform, especially compared to other options without upfront costs, had better customer service and made legal compliance easier. Jan McDowell,[9] another House candidate from Texas, found ActBlue "easy and responsive to work with." Spencer stated that candidates who adopted the platform did not do so with the intention of altering their donor pool, but rather chose to adopt ActBlue because ActBlue was superior to products offered by competitors. In many cases, processing online contributions through ActBlue offered better value, with a predictable fee based on contributions raised.

2.2 Methods

All regressions in Tables use the two-way fixed effects (TWFE) difference-in-differences modelling strategy. TWFE are a generalization of the canonical two-period difference-in-differences estimator. Candidate-level regressions are of the following form:

$$y_{it} = \beta * \text{ActBlue candidate}_{it} + \alpha_t + \theta_i + \epsilon_{it} \tag{1}$$

In Equation 4, β represents the average treatment effect on the treated of adopting ActBlue on outcome y_{it}. I define being an ActBlue candidate$_{it}$ as a binary: 1 when candidate i in year t directs website visitors to ActBlue and 0 otherwise. The outcomes y_{it} represent a candidate's contributions by donor type, average donor ideology, and behavior-based ideology. Time and candidate fixed effects are represented by α_t and θ_i respectively. Each

[8]in conversation with the author

[9]in conversation with the author

observation is a general election House Democratic candidate, for each electoral cycle in the period of study (2002-2016).

Tables in the main body also feature up to three additional specifications, space permitting. In order to evaluate pre-trending, I add in a lead term λ on ActBlue adoption. If there is little pre-trending, then the lead term will be indistinct from zero. Because non-website based ActBlue adoption is a occurs, I also include a dummy "FEC" for periods in which the candidate reports money from ActBlue in their Federal Election Commission forms but do not use ActBlue on their website. This helps distinguish the lead term from uncertainty arising from a candidate's true timing of ActBlue adoption.

$$y_{it} = \beta * \text{ActBlue candidate}_{it} + \lambda * \text{ActBlue candidate}_{i(t-1)} + \text{FEC}_{it} + \alpha_t + \theta_i + \epsilon_{it} \quad (2)$$

While I include candidate fixed effects θ_i, some candidate attributes may also vary over time. For example, we might expect candidate who goes from a very Democratic district to a lean-Republican district, via redistricting or over-time shifts in district preferences, to moderate their legislative behavior. I include controls on district safety and incumbency:

$$y_{it} = \beta * \text{ActBlue candidate}_{it} + \text{District Safety}_{it} + \text{Incumbency}_{it} + \text{FEC}_{it} + \alpha_t + \theta_i + \epsilon_{it} \quad (3)$$

Finally, I use candidate-specific linear time trends (CLST). This enables each candidate to generate linear trends in behavior over time, while standard TWFE assumes a static (slope zero) fixed effect for each candidate. For example, if a wholly responsive candidate's district becomes more conservative over time, this behavior can be accounted for by $\gamma_i * t$:

$$y_{it} = \beta * \text{ActBlue candidate}_{it} + \alpha_t + \theta_i + \gamma_i * t + \epsilon_{it} \quad (4)$$

Taken together, these specifications allow for the evaluation of the robustness and plausibility of the assumptions underlying the TWFE strategy, which are further discussed below.

2.3 Assumptions and interpretation

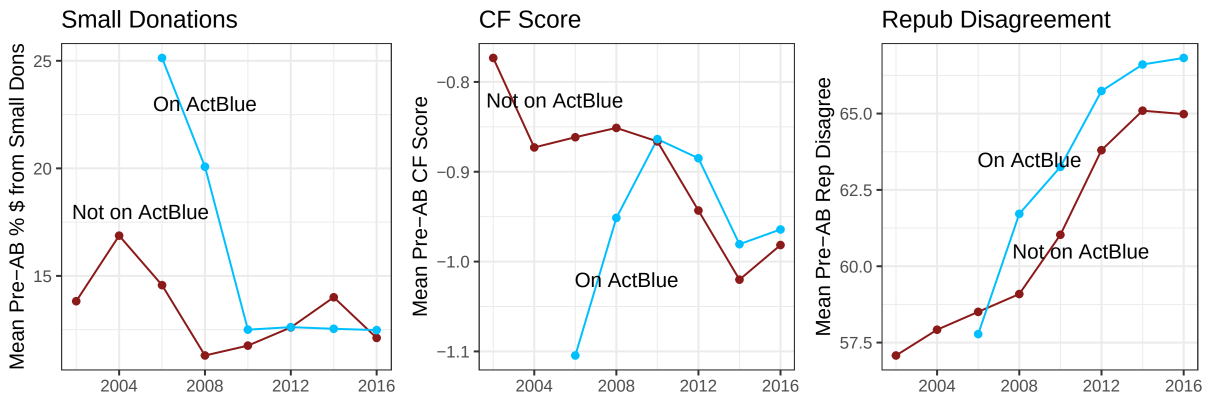
This subsection discusses methodological assumptions underlying the TWFE method and concerns on endogeneity. I take these potential threats to inference seriously. In this paper, I use a within-candidate difference-in-differences design, so that treatment effects already take into account candidate-invariant traits as well as time trends, but concerns may still arise regarding when a candidate adopts the fundraising platform and who gets excluded within the TWFE process.

2.3.1 ActBlue adoption and endogeneity

Because of the established relationship in between smaller donors and candidate extremism, a key concern in interpreting the Average Treatment Effects on the Treated as true population-wide Average Treatment Effects is that ActBlue adoption is driven by extremity-prone candidates joining the platform. While my conversations with candidates and staffers did not cite ideology as a reason for adopting ActBlue, it is true that candidates with progressive credentials were earlier adopters: in 2008, only 24 ActBlue-using legislators were elected, including nine legislators with (past or future) Congressional Progressive Caucus membership. Figure 2 shows that in ActBlue's earliest years, the type of candidates who joined ActBlue had more liberal CF Scores and higher levels of small donors prior to adoption. Candidates who select into ActBlue by 2016 always have similar or higher levels of disagreement with the median House Republican in roll call relative to non-adopters.

Since selection into ActBlue skews liberal, one concern is that results are driven by liberal legislators becoming more liberal. A similar concern is that liberal and moderate candidates may follow different trajectories over time, leading to a violation in the parallel trends assumption. Appendix Table A4 classifies candidates as "moderate" or "progressive"

Figure 2 – Donor pool characteristics of early ActBlue adopters are different from later adopters.



based on pre-ActBlue attributes, and allows for moderates and progressives to develop their own separate time fixed effects. I find that the effects in this paper are not about very liberal candidates becoming even more liberal. Candidates who score in the "moderate" half of pre-ActBlue attributes tend to have somewhat larger treatment effects for ActBlue adoption.

In-text estimates could rely too heavily on early adopters, who may have been the most enthusiastic about activating and being responsive to small donors. If results are strong among early adopters and weak among late adopters, the results may still be significant, on average. In Appendix Figure A5, I break results out by the year a candidate first uses ActBlue and find that coefficient estimates are generally similar across adoption years. Though ActBlue adoption is correlated with ideology, results are not driven by selection into ActBlue by early or extreme candidates.

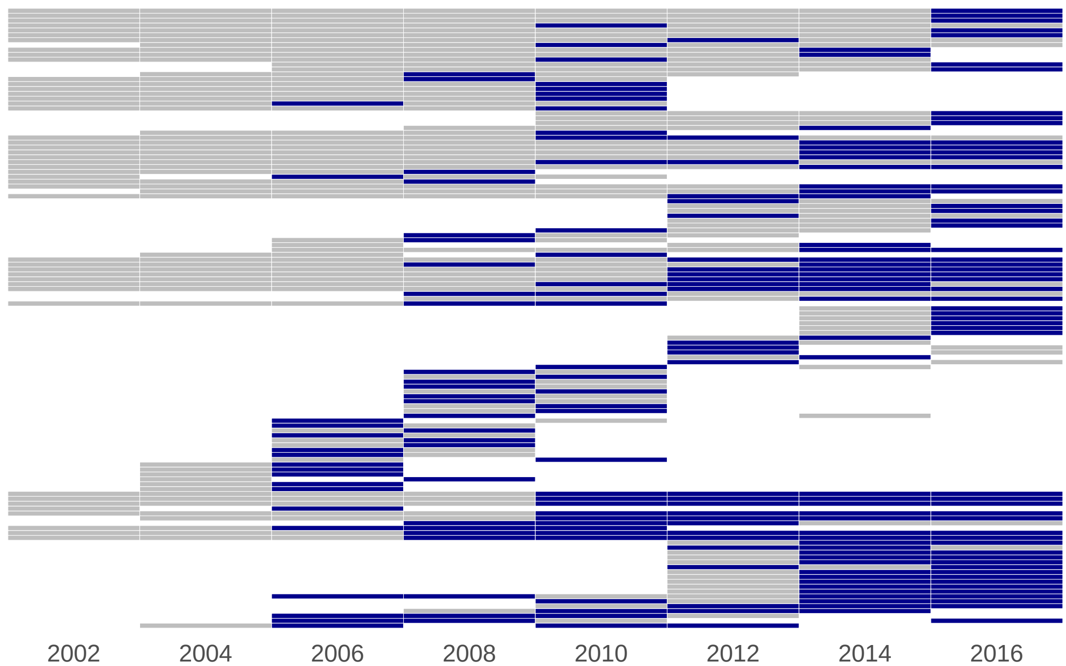
An additional potential issue is that this study covers a significant period of time – eight congressional cycles – and that those who chose to refrain from ActBlue adoption throughout this entire period of study did so strategically, and therefore may also follow different trajectories than candidates who did eventually join the platform. Appendix Table A5 drops candidates who never make the switch and again recovers coefficients similar to those in the results section.

Finally, electoral competition could induce ActBlue usage. For example, Democratic incumbents redistricted into competitive districts might take the task of fundraising more seriously and prefer ActBlue's modern interface and research into strategies to maximize contributions. If this were the case, results would not necessarily reflect the effect of ActBlue adoption. Table A7 considers the case of 2010-2012 congressional redistricting and finds that House candidates are not particularly more likely to join ActBlue after being drawn into competitive districts. Similarly, candidates might want to join ActBlue if they encounter pressure to move leftward, via mechanisms like primary competition (Brady et al., 2007) or redistricting (Kaslovsky & Kistner, 2024). Descriptive analyses in Table A8 show that candidates are not substantially more likely to join ActBlue when they encounter primary opponents or when primary elections are close. Table A9 shows that Democratic district safety imposed by redistricting also does not result in significant ActBlue adoption. I additionally present results on models with controls in order to evaluate effects after accounting for changes in district partisanship and incumbency.

2.3.2 Omitted Data

Since I focus on within-candidate variation, I can only recover a treatment effect on those who appear within the dataset at least twice. Candidates who only run once are disregarded by the TWFE process, as their individual fixed effect completely reflects their observed outcomes. Candidates who run at least twice, but never start using ActBlue act as control cases. Likewise, those who run at least twice and always use ActBlue aid in the development of time fixed effects, but again do not aid in the estimation of the effect of ActBlue adoption. The coefficient of interest, β is calculated based on the outcomes of candidates who have run at least twice, and who have run for office both with and without ActBlue. This amounts to 127 unique candidates, whose ActBlue usage trajectories are shown in Figure 3. The universe of candidates who have run at least twice is visualized in Appendix Figure A1.

Figure 3 – ActBlue adoption and usage over time by switchers. Gray denotes that the candidate ran in the cycle, but did not use ActBlue. Blue denotes the usage of ActBlue. White sections indicate that the candidate did not run in a general election.



Because electoral losers tend to only run once, estimates will inherently be skewed towards the effect of adopting ActBlue among incumbents, instead of the effect of adopting ActBlue among all Democratic candidates. In Appendix Tables A10, I repeat analyses with district and time instead of candidate and time fixed effects and show that results are substantively similar, though the magnitudes on amount raised are smaller. As I discuss in the Appendix, while the district-time fixed effects strategy allows for the comparison between different candidates in the same district, selection into the treatment becomes a larger concern.

2.3.3 Parallel trends and two-way fixed effects

The causal interpretation of β is based on a generalized difference-in-difference approach, which relies on the assumption of equal trends to recover the average treatment effect on the treated. For an ActBlue using candidate i, I assume that if i had not used ActBlue as their primary payment processor, then i would follow the same trends as candidates who did not use ActBlue. While the parallel trends assumption cannot be proved, it can be investigated. In main text tables, I report alternate regression specifications with candidate-specific linear time trends as well as specifications that use the leads of treatment adoption.

These specifications often corroborate results from the default TWFE specification, but do differ on occasion. Both using a lead term or candidate-specific linear trends places additional constraints on my model, which calculates the ATE using 127 switchers. The inclusion of a lead term necessarily drops data from my last year, 2016, as well as the year of candidate's first House run. The use of candidate-specific time trends necessitates that switchers be present in the data for at least 3 electoral cycles. Lead and CSLT strategies drop the effective number of switchers to 103 and 88, respectively.

The visual assessment of the parallel trends is difficult because my data is characterized by staggered adoption of an unbalanced panel of candidates, but event study-style plots can further aid in the investigation of parallel trends. Event study plots using TWFE are shown in Appendix Figure A3, and event study plots based on estimators from Liu et al. (2024) are presented in A4.

The TWFE estimator has recently received attention due to its unclear treatment of heterogeneous fixed effects. I examine alternative difference-in-differences strategies in the Appendix. Figure A5 discusses why the Goodman-Bacon (2021) estimator cannot be calculated in this case (and provides by-year treatment effects), Table A2 shows results using three estimators from Liu et al. (2024), and Table A3 presents results using the estimator from De Chaisemartin and d'Haultfoeuille (2020), and estimates are often near those shown in this manuscript.

3 Results

This section presents results in three sections. I begin by establishing that when candidates begin using ActBlue, they see the largest increases in cash from smaller donors. I then examine if candidates raise more money from ideologically extreme sources after ActBlue adoption, and are therefore becoming more "extreme" as measured by contribution-based metrics of candidate ideology. These first two findings establish that as it becomes easier and commonplace for small donors to give in congressional elections, the donor pool becomes skewed towards smaller and more extreme donors. I then ask whether this shift in the donor pool influences legislative behavior, and I find that elected legislators, particularly moderates, are more extreme in their congressional behavior post-ActBlue adoption.

3.1 Candidates raise more money from smaller donors after ActBlue adoption

ActBlue is often credited with boosting small donor fundraising, especially following Bernie Sanders' 2016 presidential campaign. But does the platform systematically increase small-dollar contributions, or is this reputation driven by a few high-profile fundraisers in high-salience races? Table 1 tests this by examining changes in logged contributions from non-large individual donors after candidates adopt ActBlue.

Table 1 shows that when a candidate start using ActBlue, they will, on average, raise about 58% ($\exp(0.46) = 1.58$) more dollars from unitemized donors than they did prior to ActBlue adoption, even after accounting for individual and time-varying factors. Most models show that candidates also raise more from donors giving \$200-\$500 and \$500-1000, but these increases are smaller than the increases from unitemized donors. A critical question is whether candidates are also receiving more from larger donors when they start using ActBlue: perhaps candidates aggressively solicit money from all sources when they join ActBlue. Table 2 examines whether candidates raise more money from large individual

Table 1 – When candidates join ActBlue, they receive more money from individual donors. Unitemized donors (under $200) have the largest increases

	Log $ from Unitemized				Log $ from $200-500				Log $ from $500-1000			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
AB Year	0.46	0.39	0.61	0.28	0.20	0.33	0.33	0.04	0.20	0.31	0.34	0.06
	(0.11)	(0.14)	(0.12)	(0.17)	(0.06)	(0.08)	(0.07)	(0.08)	(0.06)	(0.08)	(0.07)	(0.07)
Lead		0.14				0.03				0.02		
		(0.14)				(0.07)				(0.06)		
Cand FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FEC		✓	✓			✓	✓			✓	✓	
Controls			✓				✓				✓	
CSLT				✓				✓				✓
Obs	2796	1476	2727	2796	2748	1455	2700	2748	2752	1455	2704	2752

Note: Standard errors are clustered by candidate. CSLT = candidate-specific linear time trends.

donors as well as Political Action Committees when they start using the fundraising platform.

Table 2 – Candidates do not consistently raise more money from near-maximizing individual donors nor PACs when they begin using ActBlue.

	Log $ from Near Max				Log $ from PACs			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ActBlue Year	0.14	0.20	0.31	-0.06	0.01	0.02	0.16	-0.12
	(0.07)	(0.10)	(0.08)	(0.08)	(0.07)	(0.11)	(0.07)	(0.08)
Lead		0.10				-0.09		
		(0.07)				(0.07)		
Cand FE	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓	✓
FEC		✓	✓			✓	✓	
Controls			✓				✓	
CSLT				✓				✓
Num.Obs.	2655	1440	2609	2655	2651	1450	2609	2651

Note: Standard errors are clustered by candidate. CSLT = candidate-specific linear time trends.

Table 2's first four columns aggregate campaign contributions from individual donors giving 85% or more of a given year's campaign contribution limit.[10] Table 2's final four

[10]Donors must give 85%-100% of the cycle's campaign contribution limit over both the primary and general

columns considers funds originating from Political Action Committees. I find that candidates are not consistently receiving substantial increases in contributions from larger donors when they begin using ActBlue. It is clear from Tables 1 and 2 that campaign fundraising gains after ActBlue adoption are predominantly driven by smaller individual donors, and gains are largest for unitemized donors. Given that I find differential changes in contributions by donor type, I next consider whether small and individual donors actually account for a larger proportion of funds within a candidate's fundraising portfolio in Table 3.

Table 3 – After ActBlue adoption, the smallest individual donors comprise of a larger share of candidate's fundraising portfolios.

	% Unitemized	% \$200-500	% \$500-1000	% Near Max	% Committee
	(1)	(2)	(3)	(4)	(5)
ActBlue Year	3.83	0.56	0.59	0.29	-1.32
	(0.97)	(0.36)	(0.37)	(0.76)	(1.74)
Cand FE	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓
Obs	2832	2832	2832	2832	2832
Mean $\bar{y}$ (%)	14.21	10.10	10.61	17.75	33.65

Note: Standard errors are clustered by candidate.

Table 3 reports the aggregate percentage of contributions by donor category. When candidates join ActBlue, individual donors consistently account for a larger share of a candidate's funds. When a candidate starts using ActBlue, their share of money from unitemized donors increases by about 3.8 percentage points. Changes for all other donor types are substantially smaller. Smaller donors are indeed responsive to changes in candidate fundraising technology. When it becomes easier to donate, smaller donors give more to congressional campaigns relative to other types of donors, thereby playing a larger part in congressional fundraising. ActBlue adoption is clearly one reason for the rise of small donors in Democratic congressional fundraising.

election. For example, 2002's contribution limit was \$1000 per election (primary and general elections are separate), so to be counted, a donor must give at least 0.85*(1000+1000)=\$1700.

3.2 Candidate donor pools shift left after ActBlue adoption

To understand whether legislators are responsive to smaller donors, it is imperative to understand the preferences of donors. Given that the balance of donors shifts when candidates join ActBlue, a straightforward question is whether the ideological lean of the donorate also changes. Since smaller donors are typically more extreme than other types of donors, I anticipate these ideological shifts will be to the political left. If donors shift to the left, then ActBlue adoption may lead to ideological- and partisan-polarized behavior once legislators are in office as they are accountable to their contributors.

The only existing measurements of candidate ideology throughout the period of study are derived from campaign contributions, and operate under the assumption that a candidate's ideology is revealed by the average ideology of their contributors. In this section, I set aside the assumption of revealed preferences, and rather examine whether candidates raise money from more extreme sources when they start using ActBlue. I focus on two outcome variables traditionally used to measure candidate extremity. Dynamic CF Scores differ for each candidate each year and represent a candidate's ideology in the CF Score common space (Bonica, 2014). I generate a second outcome using contributions data with a methodology similar to Hall (2015) and Hall and Snyder Jr (2015). This procedure imputes candidate DW-NOMINATE by using overlapping donors. I refer to these scores as H-S scalings. I outline how H-S scalings are calculated, how my procedure slightly differs from Hall and Snyder Jr (2015), and discuss researcher degrees of freedom in Table A11. The H-S procedure is ideal because I can control what types of contributions are used in calculating donor extremity scores, and this flexibility allows me to investigate what types of donors are responsible for the underlying over-time shifts in donor ideology in Appendix Figure A7. For results in Table 4, H-S Scores are based on the contributory behavior of all donors who have given to at least 3 candidates, and all candidates who have received at least 10 contributions from these donors.

Table 4 – When candidates start using ActBlue, they raise money from more liberal sources.

	Dynamic CF				H-S Scores			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ActBlue Year	-0.056	-0.062	-0.079	-0.023	-0.026	-0.032	-0.025	-0.026
	(0.015)	(0.021)	(0.017)	(0.021)	(0.007)	(0.009)	(0.007)	(0.009)
Lead		-0.030				-0.011		
		(0.017)				(0.006)		
Cand FE	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓	✓
FEC		✓	✓			✓	✓	
Controls			✓				✓	
CSLT				✓				✓
Observations	2801	1482	2757	2801	2459	1422	2414	2459

Note: Standard errors as clustered by congressional candidate. Both Dynamic CF Scores and the H-S scores are temporally dynamic. CSLT = Candidate-specific linear time trends.

In Table 4, I find that candidates raise money from more ideologically extreme (more liberal) sources when they start using ActBlue, both with CF Scores and the H-S scaling. The magnitudes of these results are substantial relative to the ideological distribution of congressional candidates. Relative to the dependent variable's 2004 distribution (2004 was the last year without ActBlue switchers in the data), a -0.056 Dynamic CF Score shift in 2004 represents a 20-candidate shift to the left of the median Democratic candidate, out of 370 Democratic candidates with CF Scores. Similarly, a -0.026 H-S score shift in 2004 represents a 31-candidate shift to the left of the median candidate, out of 291 candidates with H-S Scores. These changes represent 13.9% (CF Score) and 24.7% (H-S) of a standard deviation in the distribution of 2004 Democratic ideology scores. Since I use a within-candidate design, it is also useful to consider within-candidate ideology score changes. Among candidates who ran at least twice, these coefficients represent 26.9% (CF) and 25.0% (H-S) of a candidate's mean lifetime ideological range.

While the overall average ideology of donors moves to the left, it remains unclear who drives this shift. At first glance, it seems like these results could be the result of a mundane technicality: because ActBlue is a conduit, scores might shift leftwards because the contributions from unitemized donors that flow through ActBlue must be publicly reported, and can thereby enter the ideology score calculation process. While Bonica does adjust for the percentage of conduit donations while calculating CF Scores, he does not remove them. This means that the act of using ActBlue and reporting small contributions may impact CF Scores, even if donor pools are exactly the same before and after ActBlue adoption.

However, this logic does not apply to H-S score shifts, as the contributions that are used in my H-S procedure are always over the itemization threshold of $200. Because unitemized contributions are inherently unitemized prior to the rise of campaign finance conduits, I cannot compare pre-ActBlue and post-ActBlue giving among unitemized donors. I can, however, examine how committee or itemized individual donors contribute to the shifts observed in Table 4. I recalculated H-S scores, using only certain subsets of donors in Figure A7. All donor types shift leftward.

Interpreting shifts in aggregate donor ideology requires accounting for both the composition of the donor pool (Table 3) and the ideological profile of each donor group (Figure A7). Post-adoption, small individual donors make up a larger share of contributions, and their average ideology also shifts leftward. Committees both become a smaller share of the donorate, and the remaining committee donors are also more liberal than before. The increasingly liberalness of Democratic donors is not solely due to compositional change; within each group, more liberal donors become more dominant. These findings suggest that the pursuit of contributions from smaller donors comes at the expense of moderate support.

3.3 Does congressional behavior shift after ActBlue adoption?

The results from the previous sections may not necessarily translate to a leftward shift in observed congressional behavior for three reasons. First, those with the biggest ActBlue-related ideological shifts may not win office.[11] Second, shifts in financial constituencies do not always produced aligned policy responses (Ansolabehere et al., 2003; Fouirnaies & Fowler, 2022; Fowler et al., 2020). Third, ActBlue increases the share of unitemized contributions by less than 4pp, possibly too small an effect to shift legislative behavior, especially in a within-legislator design. In this section, I test whether legislative behavior shifts to the left once a candidate joins the fundraising platform. Congressional behavior is measured in the term after ActBlue adoption: for example, if a legislator starts using ActBlue ahead of the November 2012 elections, 2013-2014 is her first "treated" term.

Measuring within-legislator behavioral change can be difficult, and very few studies have been able to provide explanations for within-legislator behavioral change. Results in this section are conditional on winning office. Not all candidates who run twice necessarily win election twice, and a comparison of Table 4 to Table 5 indicates that about 40% of candidates are lost in the transition to the legislator sample. Additionally, many standard metrics of legislator behavior, such as DW-NOMINATE, do not vary by Congress, and cannot be used when by-legislator temporal variation is necessary.[12] Other scores may allow legislators to earn new scores each session independent of their previous legislative behavior, but an underlying assumption is that over-time variation can be controlled through year fixed effects.[13] Given these caveats, Tables 5-6 examine five measures of congressional behavior.

Table 5 focuses on party unity (the rate of voting alongside the median Democrat). This dependent variable gets to the core of my argument: Democratic Members of Congress are more polarized after ActBlue adoption. Table 6 shows results for four additional dependent variables: the rate of voting against the median Republican legislator, aggregated interest group scores using scores collected by Project Vote Smart, Conservative Vote Probability (CVP) scores (Fowler, 2024; Fowler & Hall, 2012), and scores derived using IRT ideal point

[11] Appendix Table A6 replicates Tables 1-4 using only winners, and finds similar effects.

[12] I discuss Nokken-Poole DW-NOMINATE scores (Nokken & Poole, 2004) in Appendix Table A15.

[13] For example, I must assume that time-varying characteristics like partisan agenda control can be differenced out with cycle fixed effects, and do not otherwise impact the distribution of dependent variables.

estimates from Clinton et al. (2004) (abbreviated as CJR). While the development of ideology or position-taking scores is an active endeavor within the field of political science, most other new scores or scores in development either have limited time frames or are currently unavailable to the public. For the CVP, CJR, and interest group scores, a more negative score indicates a more liberal position. For the measures on party unity and Republican disagreement, a positive coefficient indicates more cohesion with the Democratic consensus and voting against the Republican consensus.

In Table 5, I analyze House roll call votes to measure how often House members vote with the Democratic majority (party unity). This measure indicates a legislator's commitment to voting with the party. I use all House votes (Lewis et al., 2023) to create this measure. In this section, I also show results that consider a legislator's pre-ActBlue ideological positioning. "Moderate" and "Progressive" ideological positioning are based on pre-ActBlue positions, with legislators with below-median values of party unity coded as "moderate," and those with above-median values of party unity coded as "progressive." This breakdown serves two purposes: it helps assess for which subsets of legislators the parallel trend assumption is most plausible, and also indicates whether moderates and progressives behave differently post-ActBlue adoption.

Table 5 shows that when a House member begins raising money through ActBlue, they begin voting alongside the majority of the Democratic party about 0.76% (pp) more often. This represents a modest increase, as the average 2004 Democrat agreed with the median Democrat on 91.6% of all bills. In the 2004 House, this coefficient corresponds to a 23-congressperson shift to the left of the median 2004 House Democrat, or about 18.5% of a standard deviation (within Congress). Of legislators who have served at least two terms in office, this coefficient accounts for about 12.7% of a legislators overall lifetime range in party unity scores. Column 2 indicates that on average, the inclusion of a lead term only slightly lowers the estimate, and the estimate from Column 4 with candidate-specific

Table 5 – When Democratic legislators use ActBlue, they exhibit higher levels of party unity.

	Party Unity (%)				
ActBlue Year	0.758	0.673	0.938	0.550	0.699
	(0.218)	(0.367)	(0.321)	(0.331)	(0.488)
Lead		0.204			0.055
		(0.332)			(0.585)
Cand FE	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓
FEC		✓	✓		✓
Controls			✓		
CSLT				✓	
Moderates only					✓
Observations	1664	1332	1646	1664	533

Note: Standard errors are clustered by candidate. In 2004 (prior to any ActBlue adoption), the mean party unity was 91.6%.

linear time trends is similar other estimates. The lead term in Column 2 is statistically insignificant, though about a third the magnitude of β, indicating that on average, there is not substantial pre-trending for Members of Congress who will soon begin using ActBlue. Column 5 examines considers only moderates, and shows that moderates exhibit slightly larger effects and much less pre-trending, indicating that the assumption of parallel trends is most plausible among moderates.

Table 6 uses four additional dependent variables that measure legislative behavior. Similar to Party Unity from Table 5, Republican Disagreement indicates the proportion of votes that a Democrat makes in opposition to the Republican majority. To obtain a measure of aggregate interest group rating, I take the first principal component from over 35 interest group ratings, which itself explains about half of all variation among Democrats in interest group ratings. These groups are listed in Appendix Table A12, and the procedure is described there in more detail. Interest group ratings are created in order to help voters make policy-informed decisions and are centered around issues of national salience, whereas other scores in Tables 5-6 consider all roll call votes. Groups will often consider position-taking and roll call votes, which helps skirt issues of agenda control and strategic roll call voting.[14]

In addition to developing scores to measure legislative behavior, I use two additional scores from the political science literature in Table 6. In general, these scores are highly correlated but are built with different assumptions and methods. Conservative Vote Probability (CVP) is a simple measure that reflects a legislator's probability of making a conservative vote relative to a fixed median legislator. The CVP method requires the manual determination of what a conservative vote entails, and I use estimates from Fowler (2024). CVP is easy to interpret: a 0.10 increase in CVP is equivalent to a 10 percentage point increase in voting conservatively, and this relationship scales linearly. The Bayesian IRT method from Clinton et al. (2004) serves as the basis for many further advancements in the measurement of ideal points beyond just Congress. I calculate CJR scores for each congressional cycle using votes from VoteView and the pscl R package, and manually set the direction of coefficients such that the Democratic mean is negative and the Republican mean is positive (as the rotation of ideal points is not identified in IRT).

Columns 1-4 of Table 6 consider how voting against the median Republican. On average, when Representatives begin using ActBlue, they exhibit a 1.45% (pp) increase in disagreement with the Republican consensus. This change represents a 26-congressperson shift to the left from the median. Taken together, the results from Table 5 and Columns 1-4 of Table 6 could represent higher unity in only low-salience situations, but results using interest group ratings convey that this is not the case. Interest groups, who evaluate Members of Congress based both on voting and position-taking on specific issues, assign legislators more extreme ratings when House members begin using ActBlue. Columns 5-8 of Table 6 show that interest group ratings do substantially shift when candidates begin using ActBlue. The 0.32 average shift leftwards in 2004 represents an 8 Representative shift away from the median Democrat (of 182 House members with scores).

[14] A common critique of roll call-based scores are that they treat co-nay voting as agreement, though in practice some members of Congress vote nay for widely different reasons.

Table 6 – Effects of ActBlue adoption on legislators' ideological and partisan positioning

	Republican Disagree				Interst Grp Rating				Conserv Vote Prob				CJR			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
AB Year	1.450	0.539	1.025	1.615	-0.321	-0.380	-0.023	-0.655	-0.010	-0.003	-0.006	-0.011	-0.064	-0.052	-0.052	-0.051
	(0.460)	(0.421)	(0.712)	(0.812)	(0.126)	(0.161)	(0.105)	(0.223)	(0.003)	(0.004)	(0.005)	(0.006)	(0.033)	(0.036)	(0.041)	(0.028)
Lead		0.699		0.805		-0.187		-0.196		-0.005		-0.003		-0.032		-0.007
		(0.406)		(0.622)		(0.162)		(0.262)		(0.003)		(0.006)		(0.026)		(0.032)
Cand FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FEC		✓		✓		✓		✓		✓		✓		✓		✓
CSLT			✓				✓				✓				✓	
Moders only				✓				✓				✓				✓
Observs	1664	1332	1664	538	1525	1253	1525	472	1664	1332	1664	527	1677	1335	1677	539

Note: Standard errors are clustered by candidate. "Moderate" is based on a candidate's pre-ActBlue value of the dependent variable.

Columns 9-12 consider Conservative Vote Probability scores, and the 1% decrease in CVP (within the main TWFE model) when a candidate uses ActBlue is equivalent to about a 12-legislator shift from the 2004 median House Democrat. Columns 13-16 use the Clinton, Jackman, and Rivers score and finds leftward movement equivalent to a 14-legislator shift from the median 2004 Democrat.

Although legislative behavior is typically treated as fixed during a legislator's term, the observed shifts reported in this paper are within-legislator. Such effects are rare, as many within-legislator designs yield null results (Fouirnaies & Hall, 2022; K. Kim, 2024). This makes it difficult to benchmark the size of the ActBlue effect against other drivers of ideological change. However, to give a sense of scale, the estimated effect of ActBlue adoption on CVP is similar to the effect of a congressperson's district becoming 20% (pp) more Democratic post-redistricting (Myers, 2024).

While the results in Tables 5 and 6 somewhat diverge in magnitude because they pick up on somewhat different behavior and are produced by measures with varying underlying assumptions, the consistent pro-Democrat and liberal direction of results make it clear that when legislators start using ActBlue and raising more from small donors, their own behavior in Congress shifts left. This is especially true for more moderate House Democrats (as measured by pre-ActBlue behavior), and the parallel trends assumption is most plausible among moderates as well. Previous scholarship has suggested that the link between small donors and legislative extremism occurs through legislative turnover, where more moderate candidates are replaced by extreme candidates who are aided by small donors (Barber, 2016). However, these results suggest that under a wide variety of measures, candidates are directly and immediately ideologically responsive to their donors, even as their donors change.

Potential pathways for smaller donor influence

While small donors lack the direct access to legislators that lobbyists and large donors enjoy, they can still shape legislative behavior through several pathways. Though these donors contribute smaller amounts, simply identifying as a donor has been shown to improve access to legislative staff (Kalla & Broockman, 2016). If smaller donors are more likely to reach out to legislators than constituents, then these imbalances in contact can shape legislators' beliefs on constituent support (Broockman & Skovron, 2018). Legislators are motivated by fundraising incentives, and it may also be the case that donors reward polarizing and extreme behavior.[15] While testing these pathways is outside the scope of this paper, existing work on individual-legislator interactions provides strong evidence that small donors can influence legislative behavior.

4 A Republican counterpart?

The preceding analyses focus exclusively on Democrats and ActBlue. Similar dynamics may arise with Republican fundraising through WinRed, though key differences suggest the effects may not be parallel. Both platforms serve as each party's dominant conduit and offer comparable tools for soliciting individual contributions. As such, WinRed may also increase small donor participation and influence legislative behavior.

However, important asymmetries remain. The Republican party exercises much more control over WinRed, meaning that fundraising strategies might be more coordinated on the right. WinRed was founded much later, and was first used in the 2020 congressional election. Much of the digital infrastructure that ActBlue brought to Democratic campaigns, like a low-friction donation interface for contributors, compliance tools, and fee structures without upfront costs, existed for Republican campaigns through other payment platforms

[15] While this last pathway has yet to be rigorously tested, anecdotal evidence, such as Rep. Joe Wilson's $2.7 million haul after yelling "you lie!" during Pres. Obama's State of the Union (Allen, 2009), and the difference in donor- and voter-centric speech indicates that this may be the case (S.-Y. S. Kim et al., 2023).

by 2020.

In Table 7, I present results on WinRed adoption, based on 2016-2022 electoral cycles. WinRed adoption is staggered over 2020 and 2022, and by 2022 about 62% of House candidates directed website visitors to donate on WinRed.[16]

Table 7 – The effects of Republican WinRed adoption differ from the effects of Democratic ActBlue adoption. Without a gain in the amount of money raised from small donors, ideological shifts do not follow.

	% $ Unitemized	CF Score	% Votes Against Dems	Party Unitys	CVP
	(1)	(2)	(3)	(4)	(5)
WinRed Year	0.807	0.029	−1.121	0.292	−0.018
	(1.368)	(0.015)	(0.625)	(0.501)	(0.007)
Observations	1,453	1,476	854	854	638
Candidate FE	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓

Note: Standard errors are clustered by candidate. N is lower for the CVP regression as estimates are not yet available beyond the 117th Congress/2020 electoral cycle.

Republican adoption of WinRed does not mirror the Democratic experience with ActBlue. On average, WinRed adopters raised just 0.8% more of their funds from small donors—far below the over 3% increase observed among Democrats. While WinRed donors became slightly more conservative than those giving to non-WinRed Republicans, this shift was smaller than the leftward movement among ActBlue donors. Notably, Republican legislators using WinRed voted less conservatively on average, with a nearly 2 percentage point decline in conservative vote probability. Voting alignment with Democrats also *increased* modestly, with no substantial changes in party unity.

A concern about the WinRed design is that those who did not immediately join WinRed are fundamentally different types of candidates. Early WinRed adoption is related to seeking

[16]This counting method differs from that of (S.-Y. S. Kim & Li, 2025), who count adoption by getting the first quarter a candidate reports raising money from WinRed. Totals vary because candidates can set up a profile and raise money from WinRed while still directing website visitors to a platform like Anedot.

party approval and financial support, as in 2019 the RNC was "threatening to withhold support from party candidates who refuse to use WinRed" (Isenstadt, 2019). However, this selection bias would likely result in the inflation of the already small magnitude of estimates, as those who joined have incentives to remain loyal to the party. Indeed, in Table A17 I show that those who made the switch in 2020 saw the biggest increases in donor ideology shifts, percentage of unitemized dollars, and percentage of votes with Republicans (though interestingly, also vote less against Democrats).

Because WinRed adoption did not consistently translate into raising more money from small donors, it should not come as a surprise that congressional behavior after WinRed adoption remained relatively static. In comparison to ActBlue, WinRed just may be a less effective vehicle for encouraging small contributions from prospective donors.

5 Discussion

Raising money is a crucial component of a successful modern congressional campaign, and American political campaigns have become increasingly expensive. How do political donations shape congressional behavior? One way is that political donations enable donors to support like-minded candidates, and money helps candidates win elections. Previous researchers have shown that candidates tend to raise money from individuals and groups who have similar viewpoints as themselves, so when these candidates reach office, they behave in ways that align with both their donors' and own interests.

Another way that political donations can impact congressional behavior is through lawmakers changing their legislative behavior to align with the desires of their donors. The case of corporate donations and pro-corporation legislative behavior has featured prominently within political science research and in mass media. Of course, corporations are not the only entities that give to campaigns. Increasingly larger shares of political contributions are originating from individuals and small donors, with unitemized donor cash accounting for over one-quarter of the money raised in the 2020 presidential and congressional races.[17] Through technological innovations like ActBlue and targeted internet ads, candidates are constantly making appeals to prospective individual donors. This paper asks whether lawmakers change their legislative behavior to align with the desires of their smaller donors. Since individual donors (including smaller donors) are known to be more politically extreme and partisan than the populace and electorate, I focus on ideological and partisan legislative behavior.

Because ActBlue adoption is staggered over time and because I utilize a battery of time-variant legislative behaviors, I am able to analyze legislator behavior before and after raising more money from smaller donors. Examining ActBlue adoption among general election Democratic candidates running for House seats, I find that campaign fundraising technology increases the percentage of funds that legislators raise from individual donors giving under $1000, and particularly those giving under $200. When Democratic candidates use ActBlue and earn more from smaller donors, their congressional behavior shifts to the left. When legislators use ActBlue, they begin voting more with the median Democrat, voting less with the median Republican, and they receiving increasingly liberal ratings from interest groups. The results in this paper are the first to establish that lawmakers are indeed responsive to smaller ideological donors: receiving money from small donors polarizes congressional behavior. Campaign contributions, including contributions from smaller donors, not only impact who wins office, but also directly affect how individual legislators behave.

While this paper primarily conceptualizes ActBlue as a tool that boosts fundraising from smaller donors, ActBlue and WinRed are also part of a broader suite of technological changes that have emerged in the era of internet campaigning. How well does ActBlue adoption explain recent trends in campaign finance? ActBlue does appear to be a serious driver in the increase in unitemized donor contributions. The role of unitemized donors in House Democratic congressional races has increased, with unitemized donors giving almost 20% of

[17] https://www.followthemoney.org/research/institute-reports/joint-report-reveals-record-donations-in-2020-state-and-federal-races

contributions in 2020, up from below 13% in 2010. Contextualized, the approximately 3.8% increase found in Table 3 represents about half of this increase. That said, ActBlue alone does not explain dramatic increases in the amounts of money raised within congressional campaigns. In Appendix Table A1, I find that ActBlue adoption is associated with about a 15-18% increase in cash raised, while the average Democratic House campaign raised around 70% more money between 2002 and 2016 (unadjusted for inflation).

Adopting ActBlue is associated with a leftward shift in measures of polarized behavior. Though the precision of estimates varies, most point estimates place the effect of adopting ActBlue at about a 4-10 percentile leftward shift relative to the median candidate or legislator. What do these shifts mean, over time, considering that nearly all Democratic candidates now use ActBlue? In terms of interest group scores, the mean has shifted from 1.63 to -0.55 within the period of study, so ActBlue's -0.3 effect is substantial. Similarly, Dynamic CF scores have moved from -0.78 to -1.11 over the same time span, and ActBlue's -0.06 shift again represents a meaningful component in leftward donor polarization. Democratic voting against the Republican median has shifted considerably within the period of study, from 48.5% to 68%, and ActBlue adoption explains about 7% of this shift. Notably, the estimates in this paper explain within-legislator behavioral changes, indicating that congressperson ideology is not completely static. Future work should seek explanations for both within-legislator and replacement-based legislative polarization.

This work provides important evidence related to campaign finance reforms that impact small donor participation. Campaign finance vouchers (Yorgason, 2025), donation matching (Malbin et al., 2012), full public financing (Harden & Kirkland, 2016; Kilborn & Vishwanath, 2021; Malhotra, 2008), spending limits (Fouirnaies, 2021), and contribution limits (Barber, 2016; Gulzar et al., 2021; Primo & Milyo, 2020) are all ways in which governments have tried elevate the role of smaller individual donors. Within the United States, these innovations have been proposed at all levels of government, but implementation has primarily been confined to state and local elections. Those empowered by governmental or private reforms that boost the role of small donors are very political (Yorgason, 2025), and in the case of congressional elections, are highly polarized. A critical challenge for policymakers and political scientists is to consider whether additional institutional attributes influence who participates in the campaign finance process, and further evaluate how variations in participation shape legislative behavior.

References

Albert, Z., & La Raja, R. (2020). Small Dollar Donors and the Evolving Democratic Party. *Unpublished Manuscript*.

Allen, J. (2009). "You lie!" Worth $2.7M for Wilson. *Politico*. https://www.politico.com/story/2009/10/you-lie-worth-27m-for-wilson-028368

Alvarez, R. M., Katz, J. N., & Kim, S.-Y. S. (2020). Hidden Donors: The Censoring Problem in US Federal Campaign Finance Data. *Election Law Journal: Rules, Politics, and Policy*, *19*(1), 1–18.

Ansolabehere, S., De Figueiredo, J. M., & Snyder, J. M. (2003). Why is There So Little Money in U.S. Politics? *Journal of Economic Perspectives*, *17*(1), 105–130.

Bafumi, J., & Herron, M. C. (2010). Leapfrog Representation and Extremism: A Study of American Voters and Their Members in Congress. *American Political Science Review*, *104*(3), 519–542.

Baker, A. E. (2016). Getting Short-Changed? The Impact of Outside Money on District Representation. *Social Science Quarterly*, *97*(5), 1096–1107.

Barber, M. (2016). Ideological Donors, Contribution Limits, and the Polarization of American Legislatures. *The Journal of Politics*, *78*(1), 296–310.

Barber, M., Canes-Wrone, B., & Thrower, S. (2017). Ideologically Sophisticated Donors: Which Candidates Do Individual Contributors Finance? *American Journal of Political Science*, *61*(2), 271–288.

Bonica, A. (2014). Mapping the Ideological Marketplace. *American Journal of Political Science*, *58*(2), 367–386.

Bonica, A. (2020). Why are There So Many Lawyers in Congress? *Legislative Studies Quarterly*, *45*(2), 253–289.

Bonneau, C. W., & Cann, D. M. (2011). Campaign Spending, Diminishing Marginal Returns, and Campaign Finance Restrictions in Judicial Elections. *The Journal of Politics*, *73*(4), 1267–1280.

Bouton, L., Cagé, J., Dewitte, E., & Pons, V. (2022). *Small Campaign Donors* (tech. rep.). National Bureau of Economic Research Working Paper.

Brady, D. W., Han, H., & Pope, J. C. (2007). Primary Elections and Candidate Ideology: Out of Step with the Primary Electorate? *Legislative Studies Quarterly*, *32*(1), 79–105.

Broockman, D., & Malhotra, N. (2020). What Do Partisan Donors Want? *Public Opinion Quarterly*, *84*(1), 104–118.

Broockman, D., & Skovron, C. (2018). Bias in Perceptions of Public Opinion among Political Elites. *American Political Science Review*, *112*(3), 542–563.

Canes-Wrone, B., & Miller, K. M. (2021). Out-of-District Donors and Representation in the US House. *Legislative Studies Quarterly*.

Carnes, N. (2018). *The Cash Ceiling*. Princeton University Press.

Clinton, J., Jackman, S., & Rivers, D. (2004). The Statistical Analysis of Roll Call Data. *American Political Science Review*, *98*(2), 355–370.

Culberson, T., McDonald, M. P., & Robbins, S. M. (2019). Small Donors in Congressional Elections. *American Politics Research*, *47*(5), 970–999.

De Chaisemartin, C., & d'Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American economic review*, *110*(9), 2964–2996.

Desmarais, B., La Raja, R., & Kowal, M. S. (2015). The Fates of Challengers in US House Elections: The Role of Extended Party Networks in Supporting Candidates and Shaping Electoral Outcomes. *American Journal of Political Science*, *59*(1), 194–211.

Fouirnaies, A. (2021). How Do Campaign Spending Limits Affect Elections? Evidence from the United Kingdom 1885–2019. *American Political Science Review*, *115*(2), 395–411.

Fouirnaies, A., & Fowler, A. (2022). Do Campaign Contributions Buy Favorable Policies? Evidence from the Insurance Industry. *Political Science Research and Methods*, *10*(1), 18–32.

Fouirnaies, A., & Hall, A. B. (2018). How Do Interest Groups Seek Access to Committees? *American Journal of Political Science*, *62*(1), 132–147.

Fouirnaies, A., & Hall, A. B. (2022). How do electoral incentives affect legislator behavior? evidence from us state legislatures. *American Political Science Review*, *116*(2), 662–676.

Fowler, A. (2024). Partisan Constituencies and Congressional Polarization. *Journal of Political Institutions and Political Economy*.

Fowler, A., Garro, H., & Spenkuch, J. L. (2020). Quid Pro Quo? Corporate Returns to Campaign Contributions. *The Journal of Politics*, *82*(3), 844–858.

Fowler, A., & Hall, A. (2012). Conservative Vote Probabilities: An Easier Method for the Analysis of Roll Call Data. *Available at SSRN 2120720*.

Goodman-Bacon, A. (2021). Difference-in-Differences with Variation in Treatment Timing. *Journal of Econometrics*, *225*(2), 254–277.

Grier, K., Grier, R., & Mkrtchian, G. (2023). Campaign Contributions and Roll-Call Voting in the US House of Representatives: The Case of the Sugar Industry. *American Political Science Review*, *117*(1), 340–346.

Gulzar, S., Rueda, M. R., & Ruiz, N. A. (2021). Do Campaign Contribution Limits Curb the Influence of Money in Politics? *American Journal of Political Science*.

Hall, A. B. (2015). What Happens when Extremists Win Primaries? *American Political Science Review*, *109*(1), 18–42.

Hall, A. B., & Snyder Jr, J. M. (2015). Candidate Ideology and Electoral Success. *Unpublished Manuscript*.

Harden, J. J., & Kirkland, J. H. (2016). Do Campaign Donors Influence Polarization? Evidence from Public Financing in the American States. *Legislative Studies Quarterly*, *41*(1), 119–152.

Hersh, E. (2020). Rage-Donating Only Made Democrats Feel Better. *The Atlantic*. https://www.theatlantic.com/ideas/archive/2020/11/folly-just-throwing-money-political-candidates/617074/

Isenstadt, A. (2019). GOP at War Over Fundraising. *Politico*. https://www.politico.com/story/2019/07/09/gop-fundraising-1404869

Kalla, J. L., & Broockman, D. E. (2016). Campaign Contributions Facilitate Access to Congressional Officials: A Randomized Field Experiment. *American Journal of Political Science*, *60*(3), 545–558.

Kaslovsky, J., & Kistner, M. R. (2024). Responsive Rhetoric: Evidence from Congressional Redistricting. *Legislative Studies Quarterly*.

Keena, A., & Knight-Finley, M. (2019). Are Small Donors Polarizing? A Longitudinal Study of the Senate. *Election Law Journal: Rules, Politics, and Policy*, *18*(2), 132–144.

Kilborn, M., & Vishwanath, A. (2021). Public Money Talks Too: How Public Campaign Financing Degrades Representation. *American Journal of Political Science*.

Kim, K. (2024). Lame Duck by Primary: Effects of Electoral Incentives on U.S. House Representatives. *Working Paper*.

Kim, S.-Y. S., & Li, Z. (2025). Keep Winning with WinRed? Online Fundraising Platform as the Party's Public Good. *Journal of Politics*.

Kim, S.-Y. S., Zilinsky, J., & Brew, B. (2023). Donate To Help Us Fight Back: Mobilization Rhetoric in Political Fundraising. *Party Politics*.

Kujala, J. (2020). Donors, Primary Elections, and Polarization in the United States. *American Journal of Political Science*, *64*(3), 587–602.

La Raja, R. (2014). Campaign Finance and Partisan Polarization in the United States Congress. *Duke J. Const. L. & Pub. Pol'y*, *9*, 223.

La Raja, R., & Schaffner, B. (2015). *Campaign Finance and Political Polarization: When Purists Prevail*. University of Michigan Press.

Lewis, J. B., Poole, K., Rosenthal, H., Boche, A., Rudkin, A., & Sonnet, L. (2023). Voteview: Congressional Rollcall Votes Database.

Li, Z., & DiSalvo, R. W. (2023). Can Stakeholders Mobilize Businesses for the Protection of Democracy? Evidence from the US Capitol Insurrection. *American Political Science Review*, *117*(3), 1130–1136.

Liu, L., Wang, Y., & Xu, Y. (2024). A Practical Guide to Counterfactual Estimators for Causal Inference With Time-Series Cross-Sectional Data. *American Journal of Political Science*, *68*(1), 160–176.

Malbin, M. J., Brusoe, P. W., & Glavin, B. (2012). Small Donors, Big Democracy: New York City's Matching Funds as a Model for the Nation and States. *Election Law Journal*, *11*(1), 3–20.

Malhotra, N. (2008). The Impact of Public Financing on Electoral Competition: Evidence from Arizona and Maine. *State Politics & Policy Quarterly*, *8*(3), 263–281.

Meisels, M. (2022). Everything in Moderation? The Effect of Candidate Extremism on Individual and Corporate PAC Fundraising. *Unpublished Manuscript*.

Meisels, M., Clinton, J. D., & Huber, G. A. (2022). Giving to the Extreme? Experimental Evidence on Donor Response to Candidate and District Characteristics. *British Journal of Political Science*, 1–23.

Miller, M., & Camberg, N. (2020). US House Primary Election Results (2012-2018). *Harvard Dataverse*.

Myers, A. C. (2024). How Do Legislators Adapt to New Electorates? Evidence from Redistricting in Congress and US State Legislatures. *Working Paper*.

Nokken, T. P., & Poole, K. T. (2004). Congressional Party Defection in American History. *Legislative Studies Quarterly*, *29*(4), 545–568.

Pettigrew, S., Owen, K., & Wanless, E. (2014). US House Primary Election Results (1956-2010). *Harvard Dataverse, v4. https://doi. org/10.7910/DVN/26448*.

Pildes, R. H. (2019). Participation and Polarization. *U. Pa. J. Const. L.*, *22*, 341.

Powell, E. N., & Grimmer, J. (2016). Money in Exile: Campaign Contributions and Committee Access. *Journal of Politics*, *78*(4), 974–988.

Primo, D. M., & Milyo, J. D. (2020). *Campaign Finance and American Democracy: What the Public Really Thinks and Why It Matters*. University of Chicago Press.

Reilly, A. (2019). How ActBlue, A Somerville-Based Nonprofit, Is Driving The Democrats' Small-Donor Revolution. *WGBH*. https://www.wgbh.org/news/politics/2019/10/08/how-a-somerville-based-nonprofit-is-driving-the-democrats-small-donor-revolution

Snyder Jr, J. M. (1992). Artificial Extremism in Interest Group Ratings. *Legislative Studies Quarterly*, 319–345.

Sokolove, M. (2022). The Losing Democrats Who Gobbled Up Money. *The New Republic*. https://newrepublic.com/article/165243/amy-mcgrath-losing-democrats-senate-fundraising

Stratmann, T. (1995). Campaign Contributions and Congressional Voting: Does the Timing of Contributions Matter? *The Review of Economics and Statistics*, 127–136.

Thomsen, D. (2023). Competition in Congressional Elections: Money Versus Votes. *American Political Science Review*, *117*(2), 675–691.

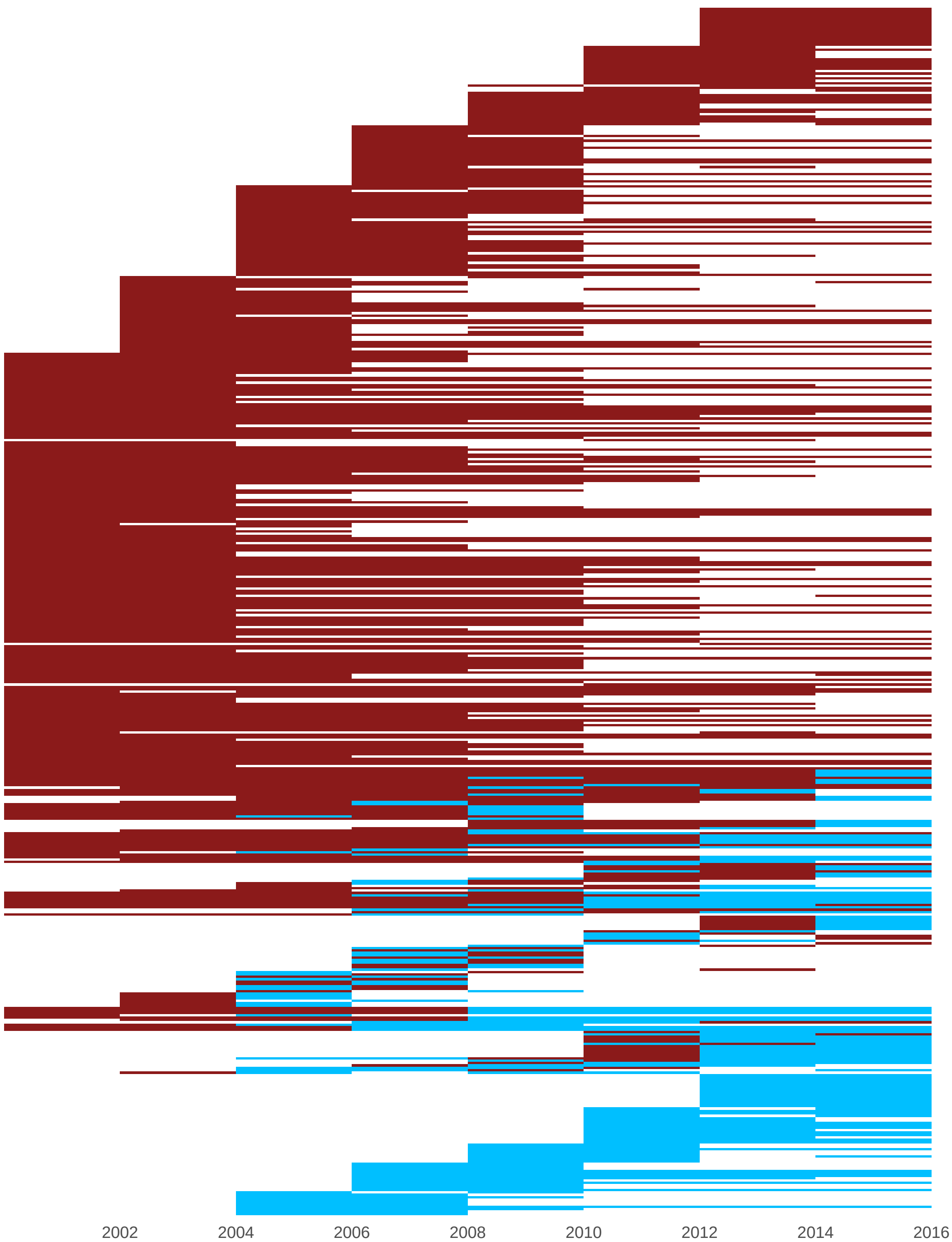
Willis, D. (2014). How ActBlue Became a Powerful Force in Fund-Raising. *New York Times*. https://www.nytimes.com/2014/10/09/upshot/how-actblue-became-a-powerful-force-in-fund-raising.html

Yorgason, C. (2025). Campaign Finance Vouchers Do Not Expand the Diversity of Donors: Evidence from Seattle. *American Political Science Review*, *119*(1), 508–516.

6 Appendix

6.1 Adoption graph for candidates who run 2 or more times

Figure A1 – Most candidates remain ActBlue users after switching on, though there is some on-off switching in the treatment. Red denotes non-ActBlue usage, and blue denotes ActBlue adoption. White means a candidate was not on the general election ballot in a given year.



6.2 Changes in campaign finance trends

In the main text, I reference but do not directly show some recent trends in the financing of congressional elections. Figure A2 visualizes trends in campaign finance for Democratic congressional candidates running in general elections, over the past two decades. Notably, the amount of money raised by candidates has drastically increased, especially since 2016. The percentage of funds from PACs has generally decreased, and remained relatively stable since 2012. The percentage of contributions from small donors has shown dramatic growth starting in 2016. The unitemized threshold has remained at $200, despite inflation.

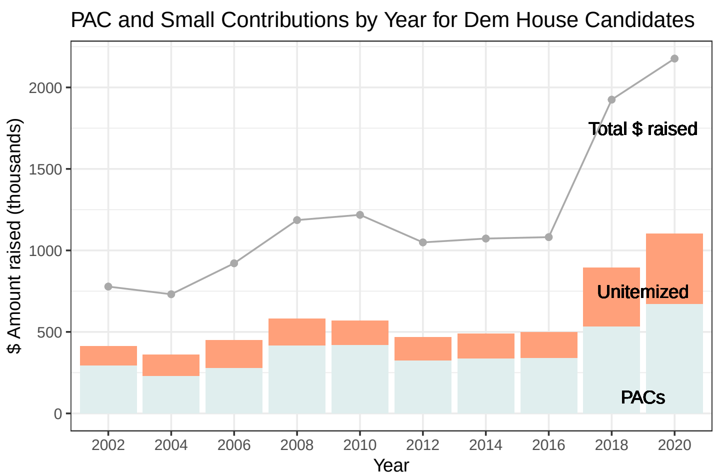


Figure A2 – Campaign contribution attributes over time.

	Log Total Raised	
	(1)	(2)
ActBlue Year	0.17	0.14
	(0.06)	(0.04)
Cand FE	✓	✓
Time FE	✓	✓
Controls	✓	
Observations	3,070	2,972

Note: Standard errors are clustered by candidate.

Table A1 – Candidates tend to raise more money when they start using ActBlue.

6.3 Event study graphs

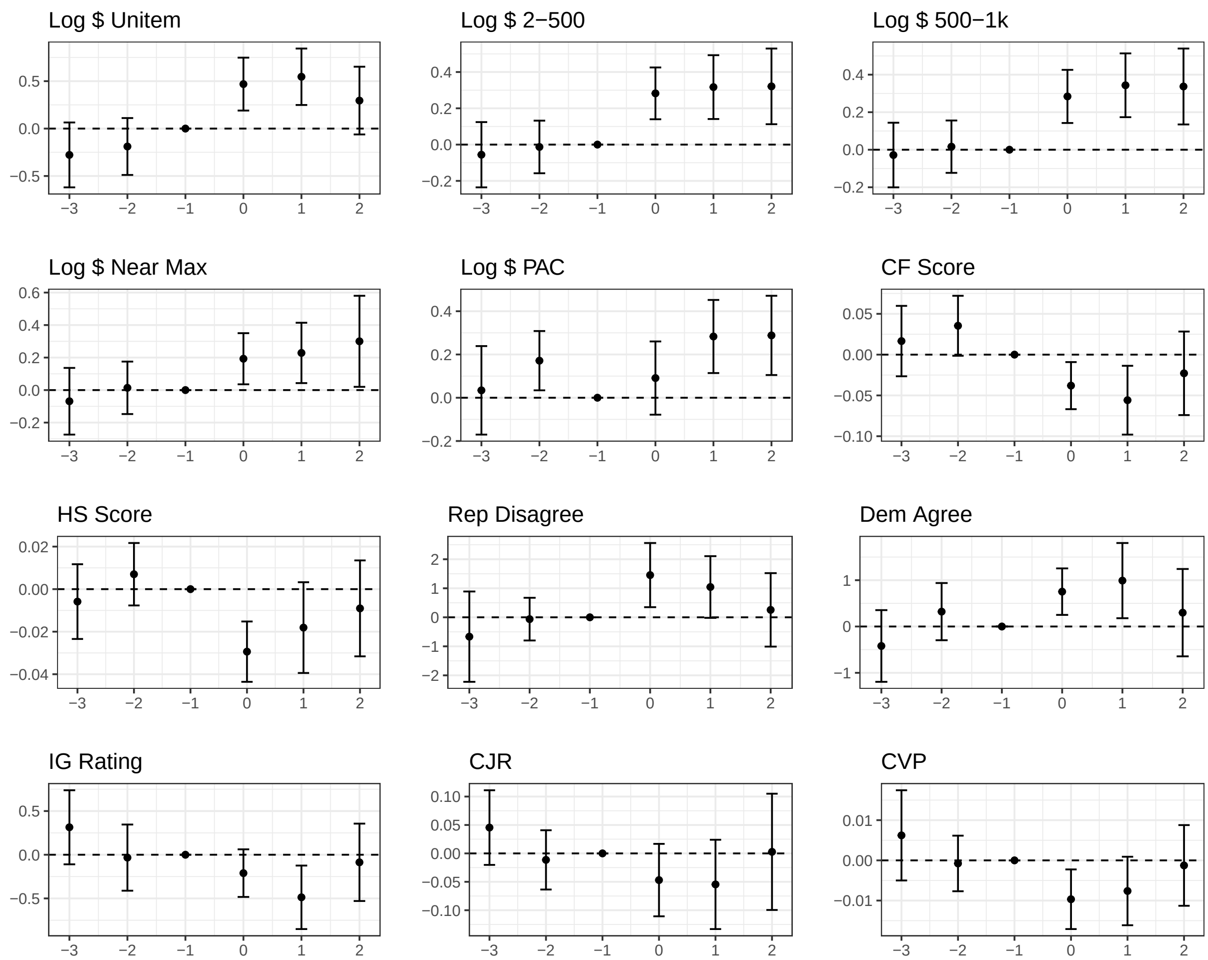


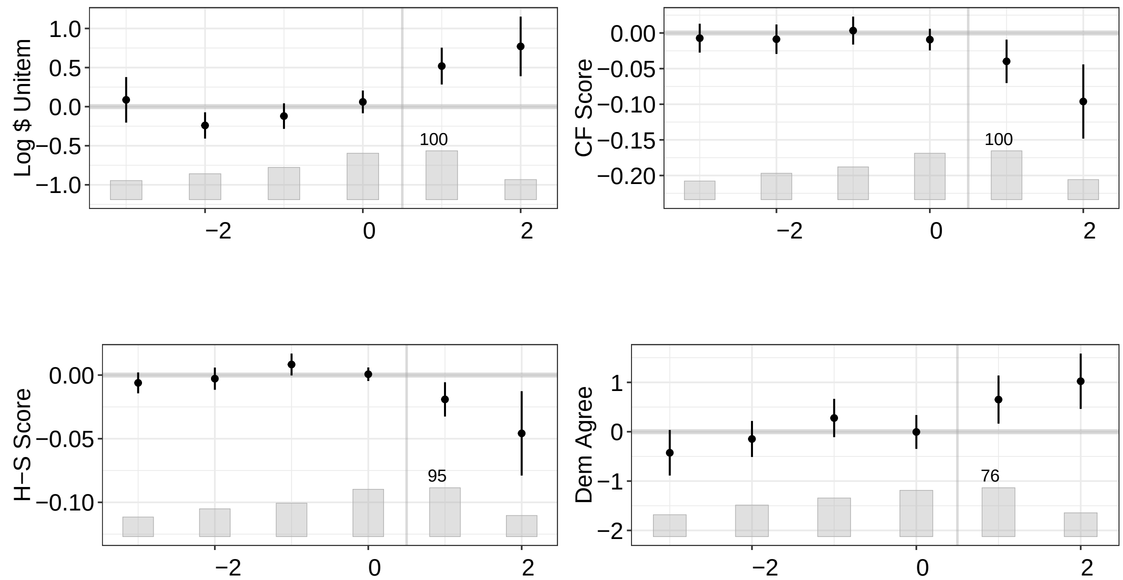
Figure A3 – TWFE event study graphs. Non-switchers are used in the development of year time trends only. X=0 denotes the first treated period.

6.4 Liu, Wang, and Xu Estimators

Table A2 – Liu, Wang, and Xu estimators by dependent variable and estimator type. FE = TWFE counterfactual (which is different from TWFE in the main text), IFE = Interactive fixed effects counterfactual, MC= Matrix completion method.

Dep Var	Estimator	Coef	SE	Dep Var	Estimator	Coef	SE
Log $ Unitem	FE	0.49	0.12	H-S Score	FE	-0.02	0.01
Log $ Unitem	IFE	0.52	0.22	H-S Score	IFE	-0.03	0.01
Log $ Unitem	MC	0.46	0.19	H-S Score	MC	-0.03	0.01
CF Score	FE	-0.06	0.02	Party Unity	FE	0.72	0.21
CF Score	IFE	-0.08	0.03	Party Unity	IFE	0.46	0.30
CF Score	MC	-0.07	0.02	Party Unity	MC	0.47	0.23

Figure A4 – Liu et al. (2024) Event-study style plots using the LWX "FE" estimator. Post-treatment effects differ from pre-treatment "effects" and are often distinct from 0. Pre-treatment exhibit no consistent trending. X=1 denotes the first treated period.



6.5 de Chaisemartin and D'Haultfœuille estimators

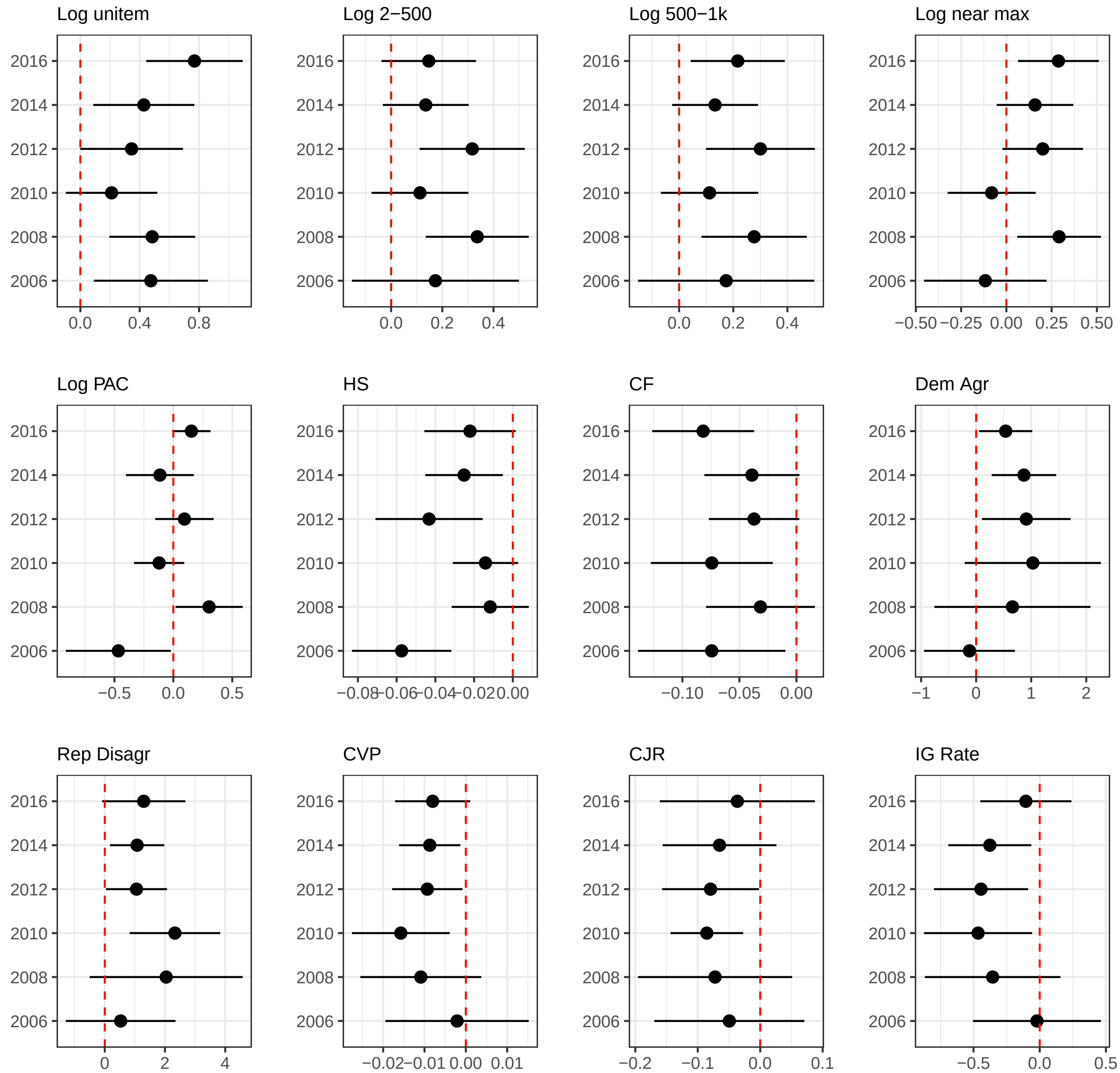
For these estimators, Placebo 1 and 2 correspond to the first and second years prior to ActBlue adoption. Year 1-3 indicate the effect within the first through third year of ActBlue adoption.

Table A3 – de Chaisemartin and D'Haultfœuille estimators

	Log Unitem	HS	CF	Party Unity	Rep Disag	CVP	CJR
Average	0.42	–0.04	–0.02	0.94	1.16	–0.01	–0.02
	(0.13)	(0.02)	(0.01)	(0.35)	(0.54)	(0.00)	(0.04)
Year 1	0.37	–0.02	–0.01	0.87	1.57	–0.01	–0.04
	(0.13)	(0.01)	(0.01)	(0.24)	(0.62)	(0.00)	(0.03)
Year 2	0.41	–0.05	–0.02	0.89	0.50	–0.01	–0.02
	(0.13)	(0.03)	(0.01)	(0.47)	(0.58)	(0.00)	(0.05)
Year 3	0.33	–0.06	–0.01	0.54	0.33	0.00	0.04
	(0.21)	(0.04)	(0.02)	(0.67)	(0.75)	(0.01)	(0.08)
Placebo 1	–0.31	0.03	0.01	0.39	–0.05	0.00	–0.02
	(0.14)	(0.02)	(0.01)	(0.33)	(0.40)	(0.00)	(0.03)
Placebo 2	–0.52	0.02	0.02	0.25	–0.06	0.00	0.06
	(0.18)	(0.03)	(0.01)	(0.51)	(1.30)	(0.01)	(0.05)
N Switch	152	194	188	151	151	151	151

6.6 Do effects vary by timing?

Figure A5 – While certain types of individuals are more likely to join ActBlue early, there is no consistent pattern wherein only late or early adopters consistently drive the effects found in the main text. Red lines are at 0.



One potential concern is that early or late ActBlue adopters are driving the effects. There are plausible stories about both early and late adopters: perhaps early adopters are young and progressive and very excited about technology that empowers small donors, or perhaps early iterations of ActBlue software were not effective. Figure 2 shows that the types of candidates who start using ActBlue in its earliest years are quite different from both non-adopters and later adopters. Only 3 incumbents joined in 2006: John Spratt, Ciro Rodriguez, and Ed Towns. None of these three had particularly high levels of small donors or CF Scores, and the extreme 2006 trends are driven by candidates who started using ActBlue as challengers. In 2008, a some candidates with very liberal CF Scores (including Mark Schauer and Hilda Solis), high levels of small donors (like Frank Pallone), progressive reputations (Marcy Kaptur), and (past/present) Congressional Progressive Caucus membership (Raul Grijalva, Andre Carson, Carol Shea Porter, Chellie Pingree, Eric Massa, Jared Polis, and others) joined the platform, both as incumbents and first-time candidates.

A model-related timing-based concern is the issue of possible negative weights under staggered adoption. Essentially, using TWFE is the result in treatment effects based on clean comparisons (untreated and treated groups) as well as forbidden comparisons (groups treated earlier against groups treated later). If there are heterogeneous treatment effect (by time) β will be mis-estimated – and β could even have the incorrect sign. To address this, the calculation of Goodman-Bacon (2021) estimators is ideal but not possible because my dataset is not a balanced panel.[18] However, I can interrogate if effects vary by timing. Figure A5 presents results. While estimates vary and standard errors are imprecise (expected given that few candidates switch in a given year), I find no consistent pattern suggesting that the results are driven by specific cohort(s).

6.7 Are there heterogenous effects based on pre-ActBlue attributes?

One concern is that the theory and evidence presented in the main paper is only applicable to a certain type of candidate. Here, I analyze by moderation/progressivism and allow for time-liberalness FE. Table A4 examines candidates classifies candidates by pre-ActBlue dependent variables (for percent unitemized, candidates who raise more from small donors are considered "progressive"). Results are similar or stronger for moderates, suggesting that effects are not concentrated on progressives.

Table A4 – Effects are typically somewhat higher, not lower, for moderate candidates.

	% $ from Unitemized	H-S Score	Dynamic CF Score	Dem Agree	Rep Disagree	Conserv Vote Prob
ActBlue Year × Moderate	5.443	-0.033	-0.070	0.658	2.706	-0.013
	(1.474)	(0.010)	(0.026)	(0.403)	(0.776)	(0.004)
ActBlue Year × Progressive	0.407	-0.012	-0.050	0.044	0.799	-0.005
	(0.428)	(0.010)	(0.016)	(0.184)	(0.908)	(0.004)
Cand FE	✓	✓	✓	✓	✓	✓
Time-Group FE	✓	✓	✓	✓	✓	✓
Num.Obs.	2832	2113	2342	1362	1362	1362

Note: Standard errors are clustered by candidate. Time-group FE allow moderates and progressives to develop separate time trends.

[18] Even with district instead of candidate FE, I cannot recover a balanced panel due to redistricting.

6.8 What if never-adopters serve as improper controls?

One concern is that candidates who never adopt ActBlue are different than those who eventually do. As I discuss pre-trends above, the parallel trends assumption assumes that after ActBlue adoption, ActBlue users would follow similar trends to non-adopters if they hadn't made the switch. Table A5 below considers that this assumption isn't true for candidates who never make the switch. Results are similar to the main text.

Table A5 – Estimates remain similar when candidates who never use ActBlue are dropped.

	Log $ from Unitem	Log $ from PAC	H-S Score	Dynamic CF Score	Dem Agree	Rep Disagree	CVP
ActBlue Year	0.418	-0.067	-0.032	-0.047	0.644	1.257	-0.009
	(0.111)	(0.067)	(0.008)	(0.015)	(0.212)	(0.436)	(0.003)
Candidate FE	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓
Drop Never Treat	✓	✓	✓	✓	✓	✓	✓
Num.Obs.	986	920	846	983	551	551	551

Note: Standard errors are clustered by candidate.

6.9 Are analyses on only winners biased?

Table A6 – Table 6 removes electoral losers from the dataset. While I cannot show how losers would vote, I find that results are very similar when losers are removed. If behavior is correlated with these measures (H-S Scores were created to estimate voting behavior of candidates), then subsetting to winners in Table 6 is unlikely to cause substantial bias.

	Log $ Unitem	Log $ 200-500	Log $ 500-1k	Log $ near max	Log $ PAC	HS Score	CF Score
ActBlue Year	0.448	0.165	0.175	0.155	-0.007	-0.025	-0.060
	(0.134)	(0.069)	(0.064)	(0.079)	(0.078)	(0.008)	(0.017)
Cand FE	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓
Num.Obs.	1634	1637	1637	1635	1634	1636	1664

Note: Standard errors are clustered by candidate.

6.10 Do candidates join AB when they expect a tight general?

Using House candidates who ran in both 2010 (2002 boundaries) and 2012 (2012 boundaries), I analyze whether anticipated competition is driving candidates' selection into Act-

Blue. Columns 1-4 of Table A7 use a TWFE DiD model to show that when candidates are re-districted into competitive districts, they do not join ActBlue at particularly higher nor lower rates. Columns 5-6 repeats the exercise with the whole set of 2010 and 2012 candidates, using state fixed effects instead of individual candidate fixed effects. Standard errors are clustered by candidate (columns 1-4) and state (columns 5-6). Under all conditions, I find that ActBlue adoption seems to be unrelated to expectations of electoral competition.

Table A7 – Being redistricted into a competitive district does not predict ActBlue entry

	Probability that a Candidate uses ActBlue					
2012 Competitive district	0.07	0.05	0.04	0.04	0.10	0.01
	(0.08)	(0.09)	(0.06)	(0.08)	(0.08)	(0.05)
Candidate FE	✓	✓	✓	✓		
State FE					✓	✓
Time FE	✓	✓	✓	✓	✓	✓
Controls		✓		✓		✓
Competitive threshold	10%	10%	15%	15%	10%	10%
Observations	354	350	354	350	774	751

Note: Competitive threshold means being districted into a congressional district where absolute difference in Obama-Romney vote choice is within 10% or 15% (such that a 55-45 race is a 10% difference).

6.11 Do candidates join ActBlue when they have to appease liberals?

When candidates expect tight primary races

Candidates might join ActBlue when they are concerned about within-party competition during primaries. If true, this could indicate that the changes observed in the main text are not the result of joining ActBlue, but rather the result of facing serious primary competition. In Table A8, I investigate whether or not candidates are more likely to join ActBlue in years where they face competition in primary elections. Candidates are not more likely to join ActBlue when their primary is close (columns 1-4). Additionally, candidates are not more likely to join ActBlue when they face any primary competitor (columns 5-6). Finally, columns 7-8 show that there is no significant relationship in between primary closeness (represented by 1 minus the primary margin, so that closer races have higher values) and ActBlue adoption. Primary results data comes from Miller and Camberg (2020) and Pettigrew et al. (2014). The number of observations is lower than in the main text because states with jungle or nonpartisan primaries (WA, CA, LA) are excluded.

Table A8 – Candidates are not more likely to join ActBlue during years where they face primary competition.

	Probability that a Candidate uses ActBlue							
Tight primary	0.03	0.03	0.03	0.03				
	(0.04)	(0.04)	(0.04)	(0.04)				
Any prim oppo					0.0003	0.0005		
					(0.02)	(0.02)		
1 - prim margin							0.01	0.01
							(0.04)	(0.04)
Candidate, Time FE	✓	✓	✓	✓	✓	✓	✓	✓
Controls		✓		✓		✓		✓
Tight race thresh.	10%	10%	15%	15%				
Observations	2,849	2,754	2,849	2,754	2,849	2,754	2,849	2,754

Note: Standard errors are clustered by candidate. The tight race threshold means that the primary is within 10% or 15% (such that a 55-45 race is a 10% difference).

When constituencies become more liberal

Similarly, candidates might want to join ActBlue when their constituency becomes more liberal. This would pose a problem if candidates both joined ActBlue and raised more money from liberal donors and behaved liberally in office because they are responsive to constituency shifts. Table A9 evaluates this possibility. Using the redistricting case, where some legislators will be newly assigned to safe Democratic districts (where the Democratic presidential candidate amasses 15%+ or 20%+ of the vote over the Republican candidate), I find null effects on ActBlue adoption. Columns 4-5 allows Democratic safety within districts to move for any reason. Causally this is a much weaker design, but I obtain similar results.

Table A9 – Candidates are not more likely to join ActBlue when their district liberalizes.

	ActBlue adoption (0/1)				
Safe Dem District	0.08	0.001		0.05	
	(0.05)	(0.07)		(0.03)	
Pres. Dem Margin			0.003		0.0003
			(0.002)		(0.001)
Candidate, Time FE	✓	✓	✓	✓	✓
Safe race threshold	15%	20%		20%	
Redist obs only	✓	✓	✓		
Observations	774	774	761	3,070	3,020

Note: Standard errors are clustered by candidate. The safe Democratic district threshold means that the most recent presidential election is beyond a 20% or 15% margin (a 55-45 race is a 10% margin).

6.12 Regressions with district fixed effects

Because of redistricting, I use congressional district-decade fixed effects. District fixed effects help overcome one issue with the main text regressions, where switching treatment status can only happen in candidates who run at least twice. Plus, over-time trends in congressional ideology and donor sources might come about when an ActBlue-using candidate replaces a legislator who never used ActBlue, and the candidate-level design cannot capture these switches. However, the use of district fixed effects raises new issues. For example, it's possible that tech-savvy candidates are both more likely to adopt ActBlue than less tech-savvy candidates who ran before them, and tech-savviness might be related to running a campaign that seeks more small online contributions, and progressivism. The directionality of coefficients is quite similar to those in the main text, and for many variables, the magnitude is similar as well. Perhaps the biggest deviations are in log dollars from PACs, which are significant and negative in Table A10, but this likely has to do with the replacement of incumbents (likely with deeper PAC connections) with political newcomers.

Table A10 – Specifications with district fixed effects (instead of individual fixed effects) for regressions in main text Tables 1-3.

	Log \$ < 200	Log \$ 2-500	Log \$ 500-1k	Log \$ near max	Log \$ PAC	HS	CF	Dem Agree	Rep Disagr	CVP	CJR	IG Rating
AB Yr	.22	.01	-.03	-.04	-.34	-.03	-.12	.65	1.25	-.01	-.07	-.28
	(.08)	(.08)	(.09)	(.09)	(.11)	(.01)	(.02)	(.25)	(.46)	(.00)	(.03)	(.11)
Dist FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Obs.	2796	2748	2752	2655	2651	2459	2801	1664	1664	1664	1677	1525

Note: Standard errors are clustered by congressional district- decade (eg CA-01, 2002-2010).

6.13 Candidate ideology based on the method of Hall and Snyder Jr (2015)

Table A11 shows how my construction of H&S scores differs from Hall and Snyder Jr (2015). The general method for constructing a Hall-Snyder score for candidate q in a given legislative term is the following. Start with a dataset of contributions and ideology/partisanship data for legislators. For each of q's donors, assign a temporary score based on a dollar-weighted average of the scores for candidates they give to (excluding q) in the same term. The H-S score for q is the dollar weighted average of the temporary scores of q's donors. Repeat for all legislators. This method is attractive because it allows for customization (choice of starting ideology scores or the imposition of contribution thresholds), but produces easy-to-calculate time-varying measures.

Table A11 – Researcher Degrees of Freedom in Generating Candidate Ideology Scores, Based on the Method in Hall and Snyder Jr (2015)

Step	H&S Procedure	My Procedure
Choose contributions	Focus on primary or all contributions and donor type	All contribs, both individual and committee
Choose ideology	Party affiliation or DW-NOMINATE	Use DW-NOMINATE
Contrib recipients	Non-incumbents or all candidates	All candidates
Contribution threshold	Only donors who make >19 distinct contributions and candidates who receive >19 distinct contributions	Make 2 or 3, receive 10 (higher similar)

6.14 Aggregating candidate interest group ratings

Table 6 uses a dependent variable of standardized legislator interest group ratings. Using a single interest group rating is often difficult: sometimes interest groups do not rate certain legislators, and interest group ratings tend to crowd around 0% or 100% (see Snyder Jr (1992)). I use the following procedure to aggregate these ratings:

1. Select interest group ratings within the applicable time period (many only rate candidates for a few years)

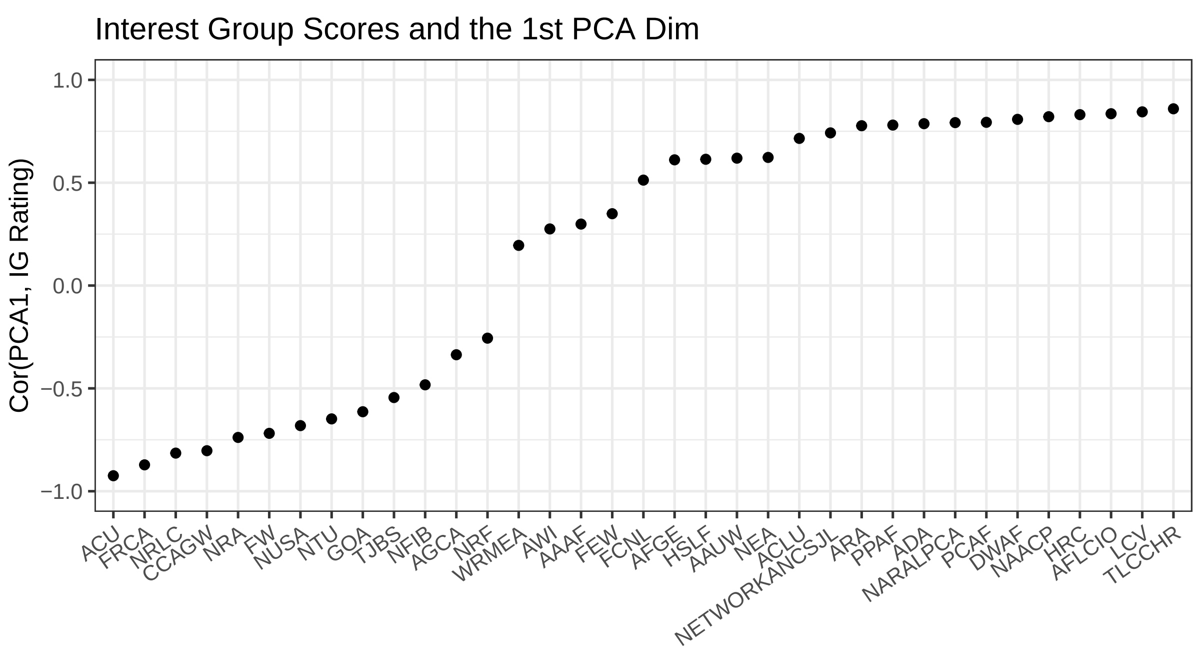
2. For each interest group i, I impute missing values by finding the interest group rating with the highest correlation (j), and predicting a legislator's i score with their j score using linear regression. This is necessary to run Principal Component Analysis without missing values.

3. Run PCA and select the first dimension (this dimension alone represents about 50% of all interest group rating variation **among Democrats** and further dimensions offer only incremental improvements)

Table A12 – List of Interest Group ratings used

Alliance for Retired Americans	Human Rights Campaign
American Association of University Women	Humane Society Legislative Fund
American Civil Liberties Union (ACLU)	League of Conservation Voters
American Conservative Union (ACU)	NARAL Pro-Choice America
American Federation of Government Employees	National Association for the Advancement of Colored People (NAACP)
American Federation of Labor and Congress of Industrial Organizations (AFL-CIO)	National Education Association (NEA)
Americans for Democratic Action (ADA)	National Federation of Independent Business (NFIB)
Americans for the Arts Action Fund	National Retail Federation (NRF)
Animal Welfare Institute	National Rifle Association (NRA)
Associated General Contractors of America	National Right to Life Committee
Council for Citizens Against Government Waste	National Taxpayers Union
Defenders of Wildlife Action Fund	NETWORK, A National Catholic Social Justice Lobby
Family Research Council (FRC) Action	NumbersUSA
Federally Employed Women	Planned Parenthood Action Fund
FreedomWorks	Population Connection Action Fund
Friends Committee on National Legislation	The John Birch Society
Gun Owners of America (GOA)	The Leadership Conference on Civil and Human Rights
Human Rights Campaign	Washington Report on Middle East Affairs (WRMEA)

Figure A6 – The first dimensions of the Interest Group PCA appears to be primarily ideological in nature. Conservative groups are bunched together, liberal groups are bunched together, and groups with primarily non-polarized and non-partisan interests have the weakest correlations with the first PCA dimension.



6.15 Does the public perceive ActBlue candidates as more liberal?

Table A13 – Using CCES data from 2006-2016, I find no change in public perceptions of local candidate ideology when candidates start using ActBlue.

	7-Pt Ideology		% Saying Cand Very Lib	
ActBlue adoption	0.01	0.03	−0.90	−1.00
	(0.04)	(0.05)	(1.05)	(1.29)
Cand FE	✓	✓	✓	✓
Time FE	✓	✓	✓	✓
Controls		✓		✓
Observations	2,328	2,259	2,328	2,259

Note: Standard errors are clustered by candidate.

CCES respondents do not perceive their local House candidates as more liberal once they begin using ActBlue. Evaluations using means of the 7-point ideology scale[19] and using the proportion of respondents who place candidates as "very liberal" remain unmoved when candidates start using ActBlue. That said, three substantial issues are present with this strategy. First, these regressions have relatively low power. The CCES starts in 2006, making it so that I cannot develop non-treated fixed effects for the first cohort of switchers. Second, the seven-point scale may not provide enough nuance to capture small changes in perceived candidate ideology. Third, public evaluations of candidates are limited by public knowledge of candidates. Nearly 46% of CCES respondents do not rate the ideology of the Democratic candidate. Even among those who do claim enough familiarity to rate candidates, familiarity and perceived ideology are not orthogonal. Less-well-known candidates are substantially more likely to be rated as moderates than well-known candidates.

6.16 Other donor types

This Table repeats similar analyses in Table 3 with different dependent variables. ActBlue adoption is not related to higher rates of funds coming from outside of state. The proportion of new donors (who haven't given to congressional races in the past two cycles) brought in is small and imprecisely measured. When candidates start using ActBlue, they do receive somewhat more from serial donors (who give to more than one congressional candidate in a cycle), though the coefficient is statistically insignificant. This graph only focuses on individual (human) itemized donors.

[19]The 2006 and 2008 CCES used a 100 point scale, and I have re-binned these to the seven point scale. Distributions look similar.

	% \$ Out of State	% New donors	% Serial donors
ActBlue Year	0.007	−0.591	0.850
	(0.861)	(1.273)	(0.858)
Candidate FE	✓	✓	✓
Time FE	✓	✓	✓
Observations	2,755	2,755	2,755

Table A14 – ActBlue adoption does not significantly increase the proportion of funds from itemized out-of-state, new donors, or serial donors.

6.17 Partisan voting and the computation of Nokken-Poole DW-NOMINATE scores

One notable omission from Table 6 are DW-NOMINATE scores, perhaps one of the most ubiquitous measures of legislator ideology within the field. NOMINATE-based scores consider who legislators share votes with (regardless of party), within the framework of a spatial model. Congresspeople receive more liberal scores when they frequently vote alongside other legislators with liberal scores. DW-NOMINATE is the predominant score from the family of NOMINATE-based scores, but cannot be used in this case as only one score is computed per legislator, even if they serve multiple terms. Nokken-Poole scores are period-specific, with fixed cutting lines established by standard DW-NOMINATE scores. Essentially, this procedure calculates by-cycle legislator scores by using boundaries drawn by stationary DW-NOMINATE scores. This procedure was initially used in order to analyze congressional behavior after party switches (a rare occurrence), and because of its reliance on fixed boundaries, it may underestimate ideological shifts if multiple legislators shift simultaneously. I present results using Nokken-Poole scores in Table A15.

A reason Nokken-Poole DW-NOMINATE scores might yield smaller results than other measures is that they were not directly designed to measure simultaneous ideological movement by multiple members of the legislature. For each legislator i, Nokken-Poole scores are calculated by letting legislators retain their "classic" DW-NOMINATE score to establish ideological boundaries, and then calculating i's by-congress ideal point "against the background of the fixed cutting lines." If multiple legislators change ideologies at once, fixed cutting lines would bias this change toward zero. In their paper, Nokken and Poole (2004) study party defections, a process which is rare and under which the assumption of one legislator's ideological change against the static ideologies of legislators is highly plausible. However, this process might be less well-suited in instances in which multiple legislators shift their ideal point simultaneously.

Table A15 highlights this issue by considering the ideological alliances (seeded on Nokken-Poole ideology scores) that candidates enter into after joining ActBlue within the second

Table A15 – When a legislator i joins ActBlue, the mean Nokken-Poole DW-NOMINATE scores of legislators who i votes with shifts to the left.

	Nokken-Poole DW	N-P Score of Voting Allies
ActBlue Year	−0.009	−0.009
	(0.006)	(0.003)
Candidate FE	✓	✓
Time FE	✓	✓
Observations	1656	1,664
'04 Leg Shift from Med	5	24

Note: Standard errors are clustered by candidate.

colummn.[20] When a legislator joins ActBlue, their voting alliances (all votes) shift to the left substantially (equivalent to a 24-legislator shift in the 2004 distribution of voting alliances). In sum, a Democratic legislator who joins ActBlue will have their average voting coalition substantially move to the left but still not have their Nokken-Poole DW-NOMINATE score move substantially to the left. Other values in Table 6 enable legislator scores to be independent each new legislative session. This "stickiness" makes it such that Nokken-Poole scores are not particularly well-calibrated for a staggered difference-in-differences design where multiple candidates adopt treatment simultaneously.

6.18 Which donor types shift to the left?

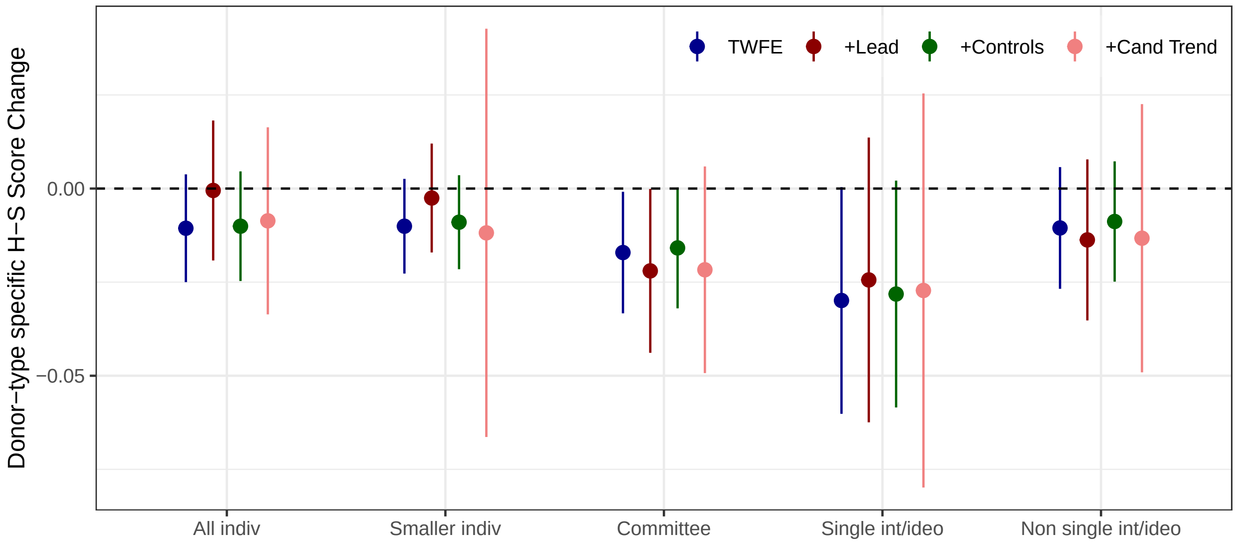
H-S scores built only on a single donor type can help me interrogate how the average ideologies of individual donors and PACs shift after ActBlue adoption. Figure A7 repeats the exercise from Table 4, but builds H-S scores by donor type.

The results in Figure A7 show that ideology scores based on all individual donors and based on only smaller individual donors shift to the left after a candidate starts using ActBlue, though at about half the magnitude found among all donors in Table 4. The values in Figure A7 exclude data from unitemized donors (whose contributions prior to ActBlue are unobservable), as well as donors who have given less than twice within a single cycle – behavior that characterizes a sizable proportion of individual contributions.

Within the literature, single interest groups and strongly ideological groups, such as the National Rifle Association or Nancy Pelosi's PAC, are conceptualized as focused on supporting candidates who are allies. Contributions coming from non-single interest and ideological sources, such as those from corporations, are often thought of as vehicles for congressional access and as ideologically moderating forces (Li & DiSalvo, 2023). The final

[20] I define voting alliances are the mean score of the legislators that vote together, where each bill is given equal weight.

Figure A7 – When a candidate joins ActBlue, they tend to raise funds from more liberal sources across multiple types of donors.



Note: Standard errors as clustered by congressional candidate. Controls are district competitiveness and incumbency status. Committee groupings as single interest/ideological are from Open Secrets, and not all committees are classified by Open Secrets. Cluster 4 uses only contributions from groups labelled as "Single Interest/Ideological," plus party committees, as labelled by Open Secrets. Cluster 5 uses all other PAC contributions, including those that are unlabelled by Open Secrets. When examining committee names, it is clear that most unlabelled groups are not single interest/ideological in nature.

three clusters in Figure A7 consider whether contributions from committees become more ideologically extreme in origin after a candidate joins ActBlue by using committee sector labels from OpenSecrets.[21]

While ActBlue-using candidates raise money from more liberal individual donors (and this fact itself can cause donor ideology scores to shift left), Figure A7 shows that candidates are also funded by more liberal committees when they start using ActBlue. This is true both for committees as a whole (third cluster), committees that are affiliated with national parties/those labeled single-interest or ideological (fourth cluster), as well as those labeled as other committee types (final cluster). The biggest changes come from party, single-interest, and ideological committees, who themselves might also be raising money on ActBlue. Though movement from other PAC sources is smaller, the effects on average donor ideology from other PAC sources are not dissimilar to average effects from individual donors. In short, when candidates start using ActBlue, they consistently raise money from

[21] To enter the regressions in Figure A7, a donor must have given as described in Table A11, appear in the DIME "Candidate/Recipient Database" and have a valid FEC ID, and be given a label by Open Secrets. Alternate matching methods that do not rely on appearing in the DIME Candidate/Recipient Database, such as using Jaro-Winkler distance to compare Open Secrets committee names with donor committee names, both introduce false positives and significantly reduce n due to false negative matches from name shortening and abbreviations.

more ideologically liberal individuals and groups.

Why does money become more liberal from *all* donor types after candidates join ActBlue? One possibility has to do with candidate messaging when they know that small donor contributions are on the table. When House candidates use ActBlue to raise funds, they often employ language that is toxic, invoking fear, disgust, and anger (S.-Y. S. Kim et al., 2023). Since campaigns may tailor messaging to maximize contributions, this polarizing language appeals to ideologues with a high propensity to give, but it may alienate more moderate individuals and committees. During the fundriasing process, campaigns may also quickly learn that ActBlue adoption brings in more cash from individual donors, and choose to take on liberal stances that appease both ideological interest groups and smaller individual donors. Alternatively, legislators may actively desire to distance themselves from more moderate or conservative donors, and realize that doing so is viable with a new source of smaller donor cash.

6.19 Robustness tables for WinRed results

This section re-runs WinRed regressions with two modifications. First, I repeat Table A5 in the Republican context. I find that when dropping never-adopters, point estimates generally decrease in magnitude. This finding suggests that part of the effect shown in the main text on WinRed adoption may be able to be attributed to non-adoption by candidates and legislators who were on a different over-time trajectory than eventual adopters.

Table A16 – This table drops Republican candidates who never adopted WinRed. Among candidates who eventually do use WinRed, the effects when they switch to WinRed are generally smaller than the point estimates recovered from the full sample.

	% $ Unitemized	CF Score	% Votes Against Dems	Party Unity	CVP
WinRed Year	0.602	0.039	−0.284	−0.402	−0.010
	(2.277)	(0.021)	(0.847)	(0.935)	(0.012)
Candidate, Time FE	✓	✓	✓	✓	✓
Observations	629	639	450	450	301

Note: Standard errors are clustered by candidate. N for CVP is lower because CVP is not yet available beyond the 117th Congress.

Next, I repeat Figure A5 in the Republican context. I show substantially larger changes for 2020 adopters than 2022 adopters. Neither of the re-analyses from Tables A16-A17 reveal results that suggest WinRed adoption exactly mirrored ActBlue adoption: in no case do I recover results near the magnitude of ActBlue results, and in the case of percent votes against Democrats, the coefficient remains in the unexpected direction.

Table A17 – Candidates who adopted WinRed in 2020 instead of 2022 have larger effects.

	% $ Unitemized	CF Score	% Votes Against Dems	% Votes w/ Rs
WR*2020	1.148	0.034	−1.757	1.053
	(1.488)	(0.016)	(0.673)	(0.641)
WR*2022	0.351	0.022	−0.292	−0.700
	(1.619)	(0.018)	(0.762)	(0.610)
Candidate, Time FE	✓	✓	✓	✓
Observations	1,453	1,476	854	854

Note: Standard errors are clustered by candidate. CVP is omitted as a dependent variable in this table as CVP scores beyond the 2020 electoral are not available.